

PROJECT REPORT

on

Design and Development of course learning and attendance Management System

Submitted in partial fulfillment of the requirements for
the award of degree

BACHELOR OF COMPUTER APPLICATIONS

of

KLE TECHNOLOGICAL UNIVERSITY

by

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SRN: 01FE23BCA205



DEPARTMENT OF COMPUTER APPLICATIONS

KLE TECHNOLOGICAL UNIVERSITY

Vidyanagar, Hubballi-580031 Karnataka.

May- 2026

**A Project Report on
Design and Development of course learning and attendance
Management System**

Modules: Attendance Management, Timetable & Scheduling, Announcements,
Analytical Reporting & Data Export

Submitted in partial fulfillment of the requirements for
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under the guidance of

Internal Guide:

Smt. Shashikala Budni

Assistant Professor,
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Applications

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Ms. Kasturi Murthy

ION IDEA, Bengaluru



**DEPARTMENT OF COMPUTER APPLICATIONS
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Vidyanagar, Hubballi-580031 Karnataka.**

May- 2026

**DEPARTMENT OF COMPUTER APPLICATIONS
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CERTIFICATE

This is to certify that the project work entitled
“Design and Development of course learning and attendance Management System”

Submitted in partial fulfillment of the requirements for
the award of degree of
**Bachelor of Computer Applications of
KLE Technological University, Hubballi, Karnataka**
is a result of the bonafide work carried out by

MS. SARVAMANGALA M C

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during the academic year 2025-2026

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ACKNOWLEDGEMENT

Every successful completion of any undertaking would be complete only after we remember and thank the almighty, the parents, the teachers, and the personalities, who directly or indirectly helped and guided during the execution of that work. The success of this work is equally attributed to all the well-wishers who have encouraged and guided throughout the execution.

I express my deepest gratitude to **Smt. Shashikala Budni** Internal Guide and **Ms. Kasturi Murthy, Mr. Nandeesh Kodipalli, and Ms. Vidya Patil, and of ION IDEA, Bengaluru**, External guide for their guidance and assistance throughout the project with great interest.

I avail this opportunity to express my deepest gratitude to **Dr. P. R. Patil, Professor & HoD** for encouraging us with his valuable tips.

I am grateful to **Dr. B. S. Anami**, Registrar for his blessings. And we are grateful to **KLE Technological University Hubballi** which has given us a bright future. I would like to thank all the people for their guidance and valuable suggestions throughout the project.

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Finally, I would like to express our sincere thanks to our **Parents and Friends** for their enormous encouragement and all others who have extended their helping hands towards the completion of our project.

Sarvamangala M C

ORGANIZATION PROFILE

Company Name: ION IDEA

Website: www.ionidea.com

Industry: Information Technology & Software Development

Location: Bengaluru, Karnataka, India

Company Type: Private Limited

About the Organization

ION IDEA is a Bengaluru-based software product and services company specializing in enterprise application development, digital transformation, and education technology solutions. The company leverages modern web technologies and agile development practices to deliver high-quality software products for clients across various industries including education, healthcare, and enterprise services. With a strong focus on innovation and quality, ION IDEA builds scalable and maintainable systems that drive measurable business impact.

Vision

To be a leading technology company that empowers organizations through innovative, scalable, and reliable software solutions that drive digital transformation and sustainable growth.

Mission

To deliver world-class software solutions using cutting-edge technologies, with a commitment to quality, customer satisfaction, and continuous innovation.

Core Service Areas

- Enterprise Web Application Development
- Education Technology (EdTech) Solutions
- Backend API Development and Integration
- Cloud Deployment and DevOps
- Digital Transformation Consulting

ORGANIZATION CERTIFICATE



May 6, 2026

To Whom It May Concern

This is to certify that **Ms. Sarvamangala** (URN: 01FE23BCA205), a student from K.L.E Technological University, was engaged with Ionidea Interactive Pvt, Ltd. as an Intern from **03-Feb-2026** to **05-May-2026**. During her internship, she actively contributed to the "Course Learning and Assessment Management application" project, demonstrating her ability to apply theoretical knowledge in practical settings.

Throughout her tenure, Ms. Sarvamangala consistently exhibited self-motivation, diligence, and a strong sense of responsibility. She took initiative in completing tasks, collaborated effectively with team members, and displayed a keen aptitude for problem-solving.

We commend her performance during the internship and extend our best wishes for her continued success in all future endeavours.

For Ionidea Interactive Pvt. Ltd.,

A handwritten signature in blue ink, appearing to read "Sangeeta", with a stylized flourish at the end.

Sangeeta Tandon
Director-Human Resources

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ABSTRACT

The increasing demand for digital transformation in educational institutions necessitates the replacement of manual record-keeping with centralized, high-performance management ecosystems. This project introduces a comprehensive Learning Management System (LMS), a data-driven platform designed to automate the academic lifecycle—from scheduling and resource sharing to student evaluation and performance analytics. By integrating a responsive React frontend with a high-speed FastAPI backend and a robust MySQL relational datastore, the system ensures real-time data synchronization and administrative precision across mission-critical modules like Communication, Timetable Logistics, and Academic Progress Tracking. The communication engine facilitates multi-channel announcements, while the Timetable module utilizes complex mapping logic to automate schedules, complemented by Attendance and Topic Coverage modules that provide granular oversight with "Finalize" states to maintain strict data integrity.

Technical implementation focuses on a decoupled architecture, leveraging Python (FastAPI) to handle complex relational joins across over 40 database tables, supporting outcomes-based education by mapping quizzes and assignments to specific Course Outcomes (CO). The frontend offers a modular interface featuring dynamic DataTables, interactive calendars, and automated report generation tools using libraries like jsPDF and XLSX. This architecture allows the system to provide stakeholders with a suite of consolidated reports—including attendance trends, registration summaries, and performance graphs—bridging the gap between raw academic data and actionable institutional insights. Ultimately, this project serves as a scalable, secure solution for modernizing educational workflows, fostering transparency, and enhancing the overall learning experience through a type-safe and responsive user interface.

CHAPTER 1

INTRODUCTION

Modern engineering education requires specialized digital systems to efficiently manage student attendance, course delivery, and academic scheduling. This project focuses on designing and developing a dedicated Course Learning and Attendance Management System. By utilizing FastAPI for the backend, React for the frontend, and MariaDB or SQL for robust data storage, the system provides a high-performance solution for tracking student engagement and learning milestones.

1.1 Literature Review

Paper 1: Martin, R. C. (2008) - Clean Code: A Handbook of Agile Software Craftsmanship

Research Paper Link:<https://www.pearson.com/en-us/subject-catalog/p/clean-code/P200000003358/9780132350884>

Summary and Interpretation:

Martin's work establishes fundamental principles for writing maintainable, readable code through meaningful naming, single-responsibility functions, proper commenting, and error handling. It emphasizes human-readable code that reduces technical debt and provides practical refactoring techniques for transforming messy code into professional-grade software.

Relevance to LMS Project:

The project applies Clean Code through modular architecture, clear API naming (e.g., `/topic_management/course_list`), and single-responsibility components. Separation of concerns between backend APIs, React frontend, and database layer reflects modularity principles. Pydantic schemas and Fast API error handling demonstrate clean code practices in educational management systems.

Limitations:

Published in 2008, it lacks modern practices like microservices, containerization, and cloud-native patterns. Examples use Java without Python-specific idioms. It doesn't cover data science, machine learning, or real-time analytics important in modern educational technology.

Paper 2: C. J. (2020) - Database Design and Relational Theory

Research Paper Link:<https://www.oreilly.com/library/view/database-design-and/9781491978466/>

Summary and Interpretation:

Date's work covers relational database design fundamentals, focusing on normalization (1NF-5NF), functional dependencies, keys, integrity constraints, and query optimization. It emphasizes

proper design for data consistency, reduced redundancy, and efficient query performance with mathematical rigor and practical applicability.

Relevance to LMS Project:

The project applies normalization through tables (IEMSCourses, CudosTopic, LMSMapInstructorTopic, TopicLessonSchedule, IEMSUsers). Proper foreign keys and CASE statements for status calculation demonstrate advanced patterns. Complex coverage status logic (green/orange/blue/red) and indexing strategies reflect Date's optimization techniques.

Limitations:

Lacks coverage of NoSQL, distributed systems, or cloud databases. Traditional relational focus may not address real-time analytics, time-series data, or document storage needs. Doesn't cover modern ORM frameworks or database-as-a-service platforms for cloud environments.

Paper 3: Pressman, R. S., & Maxim, B. R. (2020). - Software Engineering: A Practitioner's Approach.

Research Paper Link: <https://www.mheducation.com/highered/product/software-engineering-practitioners-approach-pressman-maxim/M9781260238906.html>

Summary and Interpretation:

This textbook covers the software development lifecycle from requirements through maintenance, addressing traditional and agile methodologies, architecture design, component-based design, testing, and project management. It emphasizes systematic development, quality assurance, stakeholder requirements, and frameworks for managing complexity and ensuring quality delivery.

Relevance to LMS Project:

The project follows software engineering through iterative development, NBA/NAAC requirement analysis, and component-based architecture (backend APIs, React frontend, database). API testing with curl and uvicorn configuration demonstrates testing and DevOps practices. Modular design aligns with component-based architecture principles.

Limitations:

Broad coverage lacks depth in modern technologies like FastAPI, React, or modern databases. Traditional focus may not address rapid iteration and continuous deployment common in web development. Doesn't adequately cover microservices, containerization, or cloud-native patterns. Educational domain-specific challenges receive limited coverage.

Paper 4: UNESCO. (2024) - UNESCO Guidelines for LMS Implementation

Research Paper Link: <https://unesdoc.unesco.org>

Summary and Interpretation:

UNESCO guidelines provide international LMS implementation standards focusing on accessibility, interoperability, pedagogical effectiveness, and data privacy. They emphasize

universal design, user experience, system integration, scalability, data protection, and alignment with learning theories for educational quality and equity.

Relevance to LMS Project:

The project addresses UNESCO through React frontend for cross-device accessibility, RESTful API for interoperability, and scalable normalized database. User role management and responsive design ensure accessibility. Modular architecture enables system integration, while topic-level granularity supports pedagogical effectiveness through detailed tracking.

Limitations:

Designed for diverse international contexts, may not address regional requirements or NBA/NAAC standards. General LMS focus may not cover specialized features like topic-level tracking or accreditation reporting. Lacks specific technical implementation details for FastAPI/React. Universal accessibility emphasis may conflict with specialized regulatory requirements.

5. Conclusion

This literature review demonstrates that the Institutional ERP LMS project is grounded in established software engineering principles, modern web development practices, and educational accreditation standards. The project successfully integrates concepts from clean code, software engineering methodology, FastAPI framework, database design theory, and accreditation guidelines while introducing novel features that extend beyond the scope of the reviewed references.

The system's dynamic status tracking, real-time monitoring capabilities, topic-level granularity, and accreditation-focused design represent significant advancements over traditional learning management systems. These innovations directly address the needs of educational institutions seeking to improve teaching-learning process documentation, enhance accountability, and streamline accreditation reporting.

The project's comprehensive approach, combining theoretical foundations with practical innovations, positions it as a robust solution for modern educational institutions aiming to achieve excellence in teaching quality and accreditation compliance.

1.1 Challenges/Motivation**Motivations:**

- **Centralization of Data:** Transitioning from fragmented, localized Excel files to a single source of truth in a MySQL database to ensure data consistency across departments.
- **Reduction of Manual Errors:** Automating calculations for attendance percentages and attainment marks to eliminate the high risk of human error associated with manual entry.

- **Enhanced Data Security:** Replacing open-access spreadsheets with a JWT-authenticated system featuring role-based access control to prevent unauthorized data manipulation.
- **Real-time Accessibility:** Moving away from static files that require manual sharing, enabling faculty and students to access schedules and reports instantly via a web interface.

Technical Challenges

- **Complex Data Migration:** Orchestrating the transfer of massive datasets from manual logs into over relational SQL tables while maintaining strict foreign key integrity.
- **Synchronization Logic:** Developing an API-driven layer to ensure that raw attendance and marks data accurately reflected the complex parent-child relationships found in institutional records.
- **State Management & Finalization:** Implementing a "Finalize State" mechanism to lock records once submitted, a feature impossible to enforce strictly within standard Excel sheets.
- **Legacy Format Handling:** Dealing with inconsistent data formats in historical spreadsheets (e.g., varying date formats and naming conventions) and normalizing them for the new system.

1.3 Objectives

- To design a structured database schema in MariaDB for managing students, courses, and daily attendance records.
- To develop RESTful APIs using FastAPI for high-performance data processing.
- To build responsive frontend interfaces using React for faculty to mark attendance and for students to track progress.
- To implement secure data transactions and ensure ACID compliance for critical attendance records.
- To follow clean and maintainable coding practices for the integration of all system layers.

1.4 Problem Statement

Manual attendance tracking is prone to errors and lacks real-time accessibility. Many existing systems suffer from inefficient workflow organization and weak integration between components, which reduces performance and maintainability. There is a significant need for a standalone, scalable system that uses a robust relational database like MariaDB to ensure data accuracy and smooth academic workflow handling.

CHAPTER 2

PROPOSED SYSTEM

2.1 Description of proposed system with simple block diagram

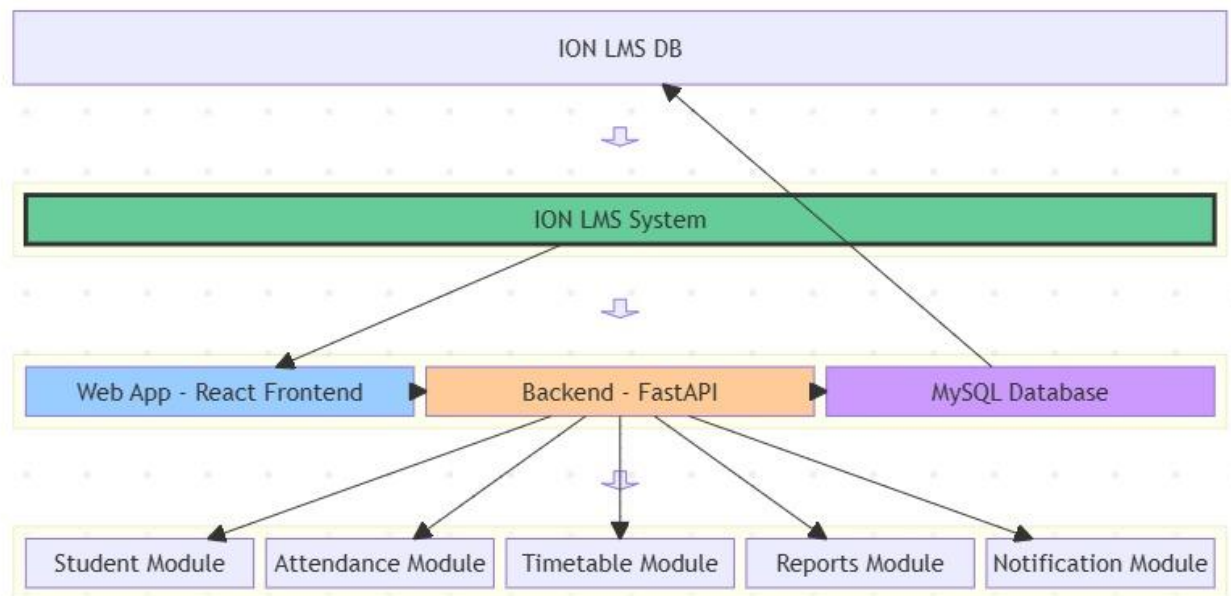


Fig 2.1 :Block diagram

1. Database Layer (ION LMS DB)

At the highest level of the hierarchy resides the global database environment. This represents the overarching data infrastructure where all institutional information is stored, serving as the primary source of truth for the entire system.

2. System Integration Layer (ION LMS System)

The green central block represents the unified system gateway. It acts as the orchestrator that manages the flow of information between the top-level database and the underlying technical stack, ensuring that the system operates as a single, cohesive unit.

3. Technology Stack (Implementation Layer)

The core of the system is built on a modern decoupled architecture:

- Web App - React Frontend: The user interface layer that handles client-side rendering and

captures user interactions.

- **Backend - FastAPI:** The high-performance API layer that processes business logic, manages authentication, and communicates with the database.
- **MySQL Database:** The relational data store specifically dedicated to handling structured data for the LMS, linked directly back to the main DB layer.

4. Functional Module Layer

The bottom tier details the specific business domains managed by the FastAPI backend. Each module is specialized to handle a distinct institutional workflow:

- **Student Module:** Manages profiles, enrollments, and academic history.
- **Attendance Module:** Handles the marking, tracking, and finalization of daily presence.
- **Timetable Module:** Controls scheduling logic and faculty/classroom allocations.
- **Reports Module:** Aggregates data for performance analytics and document generation.
- **Notification Module:** Manages real-time alerts and institutional broadcasts.

2.2 Description of Target Users

- **Students:** Primary users who interact with the system to track their academic journey, download materials, and complete assessments.
- **Faculty Members:** Operators who use the system to manage classroom logistics, evaluate student performance via Bloom's Taxonomy, and communicate with batches.
- **Administrators:** Strategic users who oversee the entire institution, ensuring syllabus coverage and generating consolidated reports for accreditation.
- **Parents:** Secondary stakeholders who can view summarized performance and attendance data to support their ward's education.

2.3 Advantages/applications of the proposed system

Advantages:

- **Data Accuracy:** Elimination of manual errors in attendance and mark calculation through automated SQL functions.
- **Performance Insights:** Moving beyond raw marks to provide cognitive level analysis .
- **Administrative Transparency:** Real-time visibility into syllabus progress and faculty scheduling.

Applications:

- **Higher Education Management:** Digitizing college and university workflows.

- **Outcome-Based Education (OBE):** Mapping student performance directly to course outcomes and cognitive levels.
- **Distance Learning Support:** Providing a unified platform for material sharing and asynchronous testing.

2.4 Scope and Limitations:

Scope:

- Can be extended to include **Biometric/IoT Integration** for automated attendance.
- Potential for **AI-driven predictive analytics** to identify students at risk of failing based on attendance trends.

Limitations:

- **Connectivity:** As a web-based React/FastAPI system, it requires stable internet access.
- **Data Entry:** The system's output is only as good as the input; it relies on faculty to timely finalize attendance and upload marks.

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

3.1 Overview of SRS

The Software Requirement Specification (SRS) document serves as a comprehensive technical blueprint for the Smart Edu-Sync platform, detailing every functional and non-functional aspect of the system. Its primary purpose is to establish a formal bridge between the institutional needs of the education management system and the technical implementation provided by the React-FastAPI stack. By centralizing all system expectations, the SRS ensures that the transition from manual, spreadsheet-based record-keeping to a digital, API-driven architecture remains consistent, secure, and aligned with the principles of Outcome-Based Education (OBE).

This document clearly defines the scope of the project, which encompasses the entire academic data lifecycle, including user authentication, curriculum mapping, attendance tracking, and attainment analytics. It outlines how the system will interact with different user personas—Admins, Faculty, and Students—while specifying the hardware and software constraints necessary for optimal performance. Furthermore, the SRS provides the criteria for system validation, ensuring that the final deployment effectively eliminates data redundancy through its MySQL relational structure and maintains high security through JWT-based authorization. Ultimately, this specification acts as the definitive guide for developers, testers, and stakeholders throughout the software development life cycle.

3.2 Requirement Specifications

3.3.1 Functional Requirements

- User Authentication: Secure login/logout and role-based access control for Students, Faculty, and Admins.
- Attendance Management: Real-time marking with a Draft/Finalized workflow to prevent data tampering.
- Academic Logistics: Automated class scheduling with a Conflict-Detection engine to prevent timetable overlaps.
- Evaluation Suite: Mapping of quiz/assignment scores to Bloom's Taxonomy for cognitive level analysis.
- Notification System: Role-based broadcasting of alerts with Seen-Flag tracking for

engagement monitoring.

- Analytical Reporting: Instant generation of consolidated marks and attendance reports by aggregating data from SQL tables.
- Real-time CRUD Feedback: Instant feedback from the MySQL database when performing create, read, update, or delete operations.

3.3.2 Use case diagrams

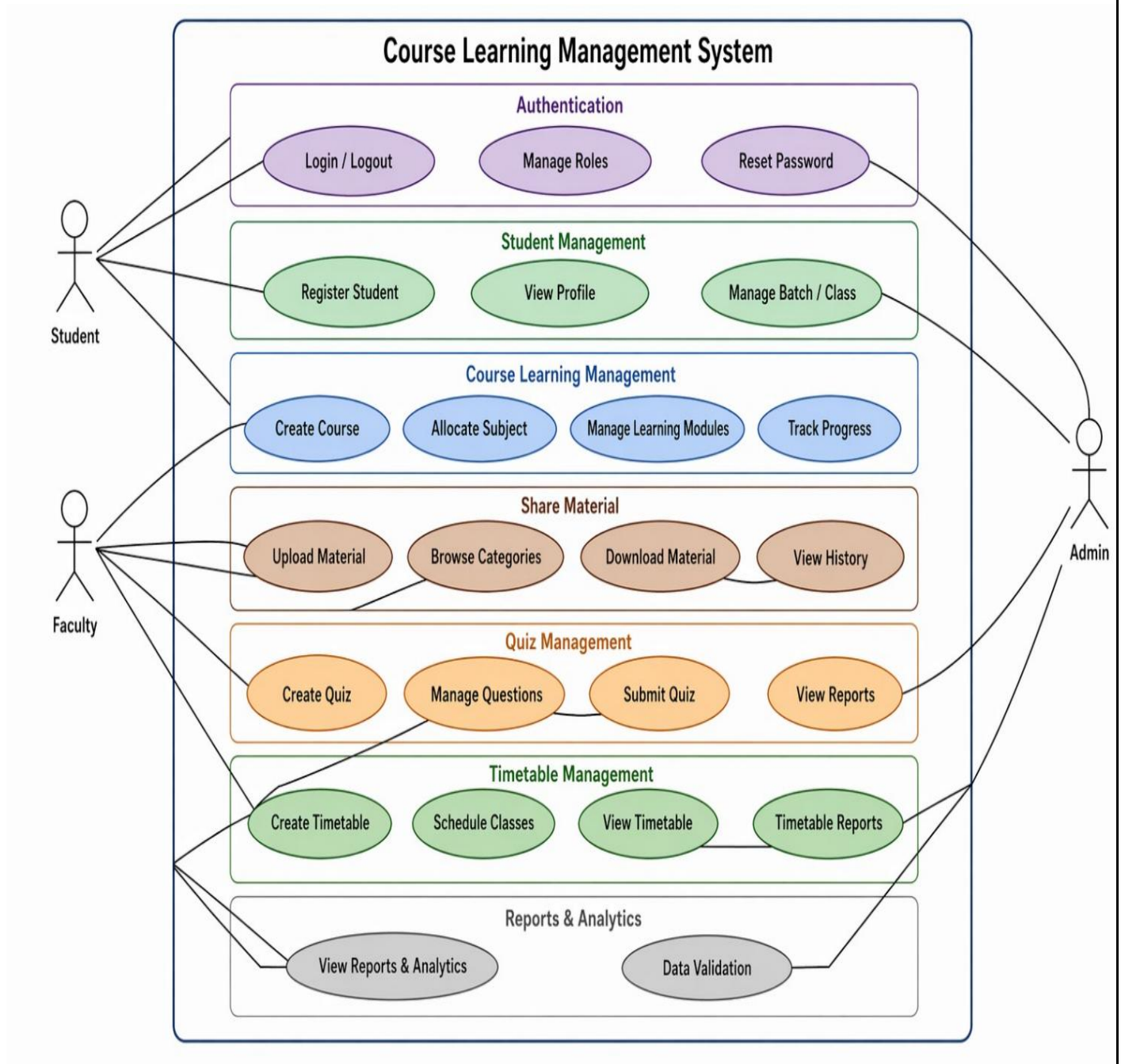


Fig 3.1 use case diagram

3.3.3 Use Case descriptions using scenarios (Pressman Template)

Module 1: Authentication & Role Management

Use Case Element	Description
Use Case ID	UC-01
Use Case Name	Manage Roles & Access
Primary Actor	Admin
Goal	To assign specific permissions and roles (Faculty/Student) to users.
Preconditions	Admin must be logged in and users must be registered in the database.
Trigger	Admin accesses the "Manage Roles" section in the dashboard.
Main Success Scenario	1. Admin selects a user from the registry. 2. Admin assigns a specific role (Faculty or Student). 3. System updates the user's JWT permissions. 4. System saves the role mapping in the users table.
Extensions	User already has role: System prompts for role update/confirmation.
Post-conditions	User gains access to role-specific protected routes and features.

Module 2: Course & Learning Management

Use Case Element	Description
Use Case ID	UC-02
Use Case Name	Create and Allocate Subject
Primary Actor	Faculty
Goal	To set up a new academic course and link it to a specific batch/term.
Preconditions	Faculty must have curriculum management permissions.
Trigger	Faculty clicks on "Create Course" in the management dashboard.
Main Success Scenario	1. Faculty enters course name, code, and learning modules. 2. Faculty allocates the subject to a specific Batch/Class. 3. System validates that the course code is unique.

Use Case Element	Description
	4. System creates a new course entry in the MySQL database.
Extensions	3a. Duplicate code: System displays "Course Code already exists" alert.
Post-conditions	The course is live and visible to assigned students for material access.

Module 3: Share Material

Use Case Element	Description
Use Case ID	UC-03
Use Case Name	Upload and Download Learning Material
Primary Actor	Faculty (Uploader) / Student (Downloader)
Goal	To distribute academic resources (PDFs, PPTs) to a specific class.
Preconditions	Course must be allocated; User must be logged in.
Trigger	Faculty selects "Upload Material" or Student selects "Download Material."
Main Success Scenario	1. Faculty selects a category and uploads a file. 2. System stores the file in File Storage and logs the path in the DB. 3. Student browses categories and clicks "Download." 4. System fetches the file and initiates a client-side download.
Extensions	Unsupported format: System rejects file and lists allowed extensions.
Post-conditions	Resource is successfully shared and logged in the "View History" log.

Module 4: Quiz Management

Use Case Element	Description
Use Case ID	UC-04
Use Case Name	Submit and Grade Quiz
Primary Actor	Student (Submitter) / Faculty (Manager)
Goal	To conduct digital assessments and track academic results.

Use Case Element	Description
Preconditions	A quiz must be created and scheduled by the Faculty.
Trigger	Student clicks "Submit Quiz" after completing questions.
Main Success Scenario	1. Student answers all questions and submits. 2. System calculates the score based on predefined answers. 3. System records the submission time and result in the marks table. 4. Faculty views reports to track class-wide performance.
Extensions	Session timeout: System auto-submits current progress.
Post-conditions	Marks are available for attainment analytics and student profiles.

Module 5: Timetable Management

Use Case Element	Description
Use Case ID	UC-05
Use Case Name	Schedule Classes & View Timetable
Primary Actor	Faculty / Admin
Goal	To establish a conflict-free weekly schedule for a batch.
Preconditions	Subjects and Faculty members must be assigned to the batch.
Trigger	Faculty accesses the "Timetable Management" grid.
Main Success Scenario	1. Faculty selects a day and time slot. 2. Faculty assigns a subject and clicks "Save." 3. System performs a conflict check (Faculty/Room availability). 4. System publishes the schedule to the Student dashboard.
Extensions	3a. Conflict detected: System prevents saving and highlights the error.
Post-conditions	All users see the updated weekly schedule in their respective calendars.

Module 6: Reports & Analytics

Use Case Element	Description
Use Case ID	UC-06
Use Case Name	View Reports & Data Validation
Primary Actor	Admin / Faculty
Goal	To analyze institutional data and ensure database integrity.
Preconditions	Sufficient data (attendance/marks) must exist in the database.
Trigger	Admin clicks on "Data Validation" or "View Reports."
Main Scenario Success	1. Actor applies filters (Date, Batch, Course). 2. System aggregates data and generates visual charts (Recharts). 3. Admin verifies records against physical logs (Data Validation). 4. Actor exports the summary to PDF/Excel.
Extensions	4a. Empty results: "Export" button is disabled; system shows "No Data."
Post-conditions	Accurate institutional insights are generated for decision-making.

3.3.3 Nonfunctional Requirements

3.3.3.1 Performance Requirements

- Speed: API response times must be under 2 seconds for standard data fetches.
- Concurrency: Support for 100+ concurrent users without UI lag.
- Rendering: AG Grid virtualization must handle 1000+ rows at 60 FPS.

3.3.3.2 Safety Requirements

- Data Integrity: Transaction-based saves to prevent partial data loss.
- Input Validation: Strict Zod/Yup schemas to block corrupted data entries.
- Error Handling: Graceful recovery from API failures via global catch blocks.

3.3.3.3 Security Requirements

- Auth: Secure JWT-based login with token expiration.
- RBAC: Strict role-based access; Students cannot edit marks or attendance.
- Protection: CORS whitelisting and HTTPS encryption for all data transfers.

3.3.3.4 Usability

- Responsiveness: Mobile-first design using Tailwind CSS for all devices.
- Navigation: Logical UI allowing users to find any feature within 3 clicks.
- Feedback: Real-time Toastify alerts for all success/error actions.

3.3.3.5 Other (Reliability & Scalability)

- Scalability: Modular React architecture for easy addition of new features.
- Cross-Browser: Full compatibility with Chrome, Firefox, Safari, and Edge.
- Maintainability: Clear TypeScript types and ESLint standards for clean code.

3.4 Software and Hardware requirement specifications

3.4.1 Software Requirements

The software stack focuses on a React-FastAPI-MySQL architecture, ensuring high speed and type safety across the application.

- Operating System: Windows 10/11, macOS, or Linux (Ubuntu 20.04+).

- **Frontend Technologies:**

- Framework: React.js 18.3.1
- Language: TypeScript 4.9.5 (for strict type safety)
- Styling: Tailwind CSS 3.4 & Bootstrap 5
- Build Tool: Vite or React Scripts 5.0.1

- **Backend Technologies:**

- Framework: FastAPI (Python-based asynchronous framework)

- ORM: SQLAlchemy (for database mapping)
- Authentication: PyJWT (for JSON Web Token management)
- Database: MySQL 8.0+ (Relational Database Management System)
- **Development Tools:**
- IDE: Visual Studio Code
- API Testing: Postman or Insomnia
- Version Control: Git & GitHub

3.4.2 Hardware Requirements

The following hardware configurations are recommended for hosting and running the application smoothly.

A. Development/Server Side

- Processor: Intel Core i5 (8th Gen) or AMD Ryzen 5 and above.
- RAM: Minimum 8 GB (16 GB recommended for concurrent Docker/Database usage).
- Storage: 256 GB SSD (Minimum 50 GB free space for project assets and DB logs).
- Internet: Stable connection for API synchronization and dependency management.

B. Client/End-User Side

- Device: Desktop, Laptop, Tablet, or Smartphone.
- Processor: Dual-core 1.5 GHz or higher.
- RAM: Minimum 2 GB for browsers to render complex data grids (AG Grid).
- Display: Minimum resolution of 1024x768 for administrative dashboards; responsive support for mobile screens.

3.5 GUI of proposed system (snap shots) navigation from Home screen to end results

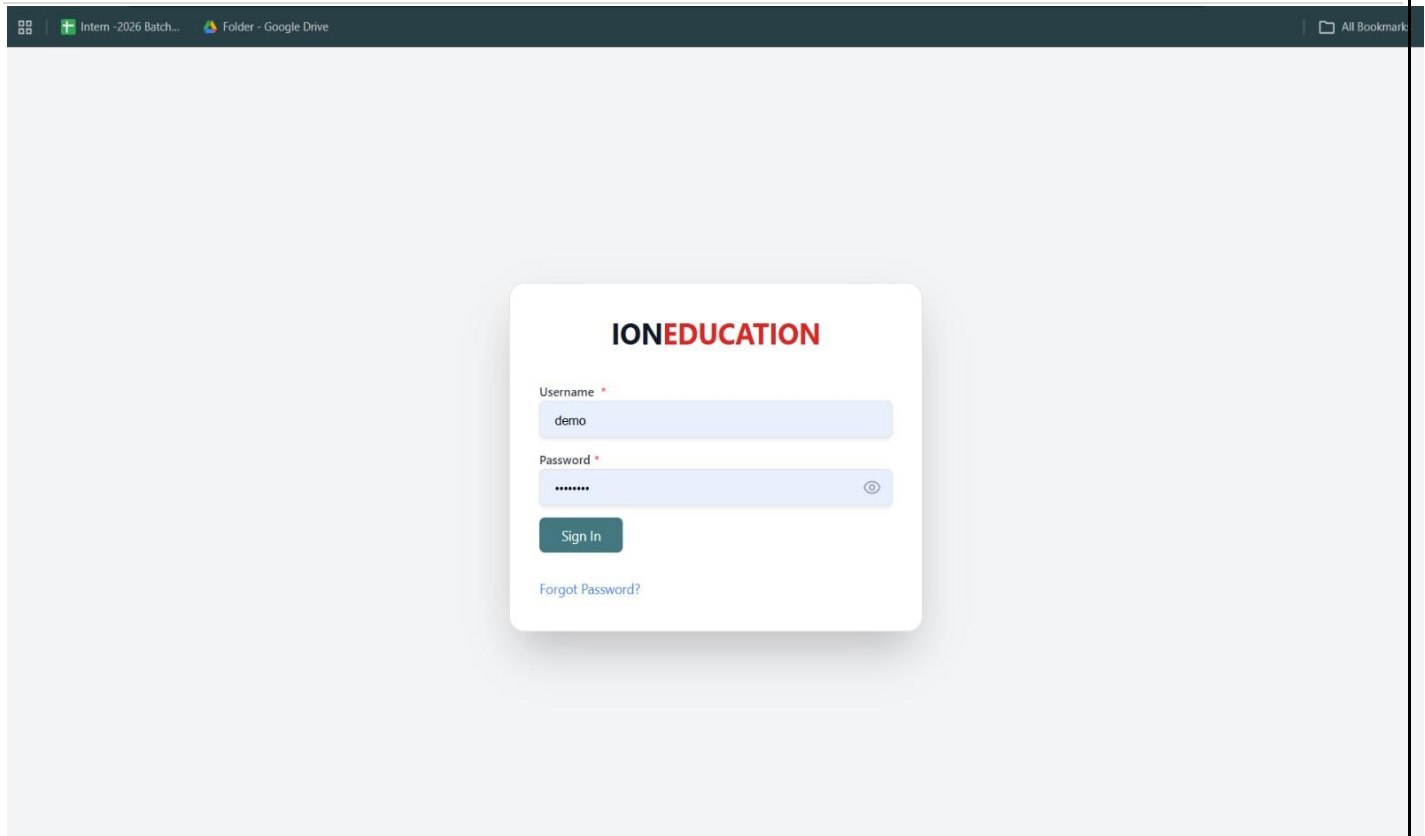


Figure 3.1: User Authentication Portal

The login interface serves as a secure gateway for the system, utilizing JWT-based authentication to verify user credentials. It ensures role-based access control by directing students, faculty, and administrators to their respective authorized dashboards

IonEMS Demo Organization

Home / Timetable

Curriculum: B.E in CSE 2024-2028 Term: 1 - Semester Section: A Timetable: New Timetable Start Date: 14-04-2026 End Date: 20-04-2026

Start Time: 06:00 End Time: 10:00 Regular/Bypass Method: Regular

[Schedule Class](#) [Options](#)

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
06:00 AM							
07:00 AM							
08:00 AM	PHY101 08:00 AM - 12:00 PM	PHY101 08:00 AM - 12:00 PM					
09:00 AM	PHY101 08:00 AM - 12:00 PM	PHY101 08:00 AM - 12:00 PM		DS101 09:00 AM - 10:00 AM			

Demo User Change Password

Figure 3.2: Timetable Configuration Portal

This portal allows administrators and faculty to design and manage the weekly academic schedule. It includes selection filters for Curriculum, Term, and Section, alongside date range pickers. The dynamic grid automatically populates scheduled classes (e.g., PHY101, DS101) into their respective time slots, providing a visual representation of the institution's resource allocation for the week.

Figure 3.3: Schedule Class Modal

This management interface allows administrators to define specific day-wise time slots for a course, such as mapping CHE101 to Wednesday and Saturday sessions. It captures precise start and end times while validating the entry against curriculum parameters like academic batch and section ID.

Mon	Tue	Wed	Thu	Fri	Sat	Sun
30	31	1 MATH101 09:00 — 10:00	2 PROG101 10:15 — 11:30	3	4 MATH101 10:00 — 01:00 MATH101 10:00 — 12:00	5
6 MATH101 10:00 — 01:00 MATH101 10:00 — 12:00	7 MATH101 10:00 — 01:00 MATH101 10:00 — 12:00	8 MATH101 10:00 — 12:00	9 MATH101 10:00 — 12:00	10 MATH101 10:00 — 12:00	11	12
13 PROG101 09:00 — 11:00	14 PHY101 08:00 — 12:00 PHY101 10:15 — 12:30	15 PROG101 09:00 — 11:00	16 DS101 09:00 — 10:00	17	18	19
20 PHY101 08:00 — 12:00	21	22	23	24	25 Today	26

Figure 3.4: Timetable Configuration Portal

This portal allows administrators and faculty to design and manage the weekly academic schedule. It includes selection filters for Curriculum, Term, and Section, alongside date range pickers. The dynamic grid automatically populates scheduled classes (e.g., PHY101, DS101) into their respective time slots, providing a visual representation of the institution's resource allocation for the week.

Student Attendance Report

Curriculum * Term * Course * Section * From Date *

B.E in CSE 2024-2028 1 MATH101 A 01-01-2025

To Date *

10-04-2026

☐ Only Present Dates

Search / Generate

Available lesson dates: 2025-01-01 to 2026-04-10

USN	Student Name	Present	Absent
1BM24CS001	Aarav Sharma	1	0
1BM24CS002	Ananya Shetty	0	1
1BM24CS005	Karthik Reddy	0	1
1BM24CS004	Priya Nair	1	0
1BM24CS003	Rohan Kulkarni	1	0

Demo User
Change Password

Figure 3.5: Student Attendance Report

This reporting interface generates a detailed list of "Present" and "Absent" counts for each student based on a selected date range and curriculum. It is a critical administrative tool used to verify attendance eligibility for final internal assessments and university examinations.

Consolidated Student Marks Report

Department * Curriculum * Term * Section *

Computer Science & Engineering B.E in CSE 2024-2028 1 - Semester A

Course

MATH101 - MATH101 X PHY101 - PHY101 X CHE101 - CHE101 X

PROG101 - PROG101 X DS101 - DS101 X

All courses selected

☒ Include Total Marks

Custom Range Export Generate Report

Consolidated Student Marks Report Consolidated Student Marks Graph

CURRICULUM TERM SECTION SELECTED COURSES INCLUDE TOTAL

B.E in CSE 2024-2028 1 - Semester A 5 Yes

Sl. No	USN	Student Name	MATH101 MATH101	PHY101 PHY101	CHE101 CHE101	PROG101 PROG101	DS101 DS101	CIA Total
1	1BM21CS001	Unknown Student (1BM21CS001) Identity fallback		60				
2	1BM21CS002	Unknown Student (1BM21CS002) Identity fallback		70				

Demo User
Change Password

Figure 3.6: Consolidated Student Marks Report

This interface provides a detailed tabular view of student performance across multiple subjects. It includes cascading filters for Department, Curriculum, Term, and Section, alongside a multi-select Course field. The generated grid displays student USNs, names, and their corresponding Continuous Internal Assessment (CIA) totals, with built-in functionality for custom date ranges and data exportation.

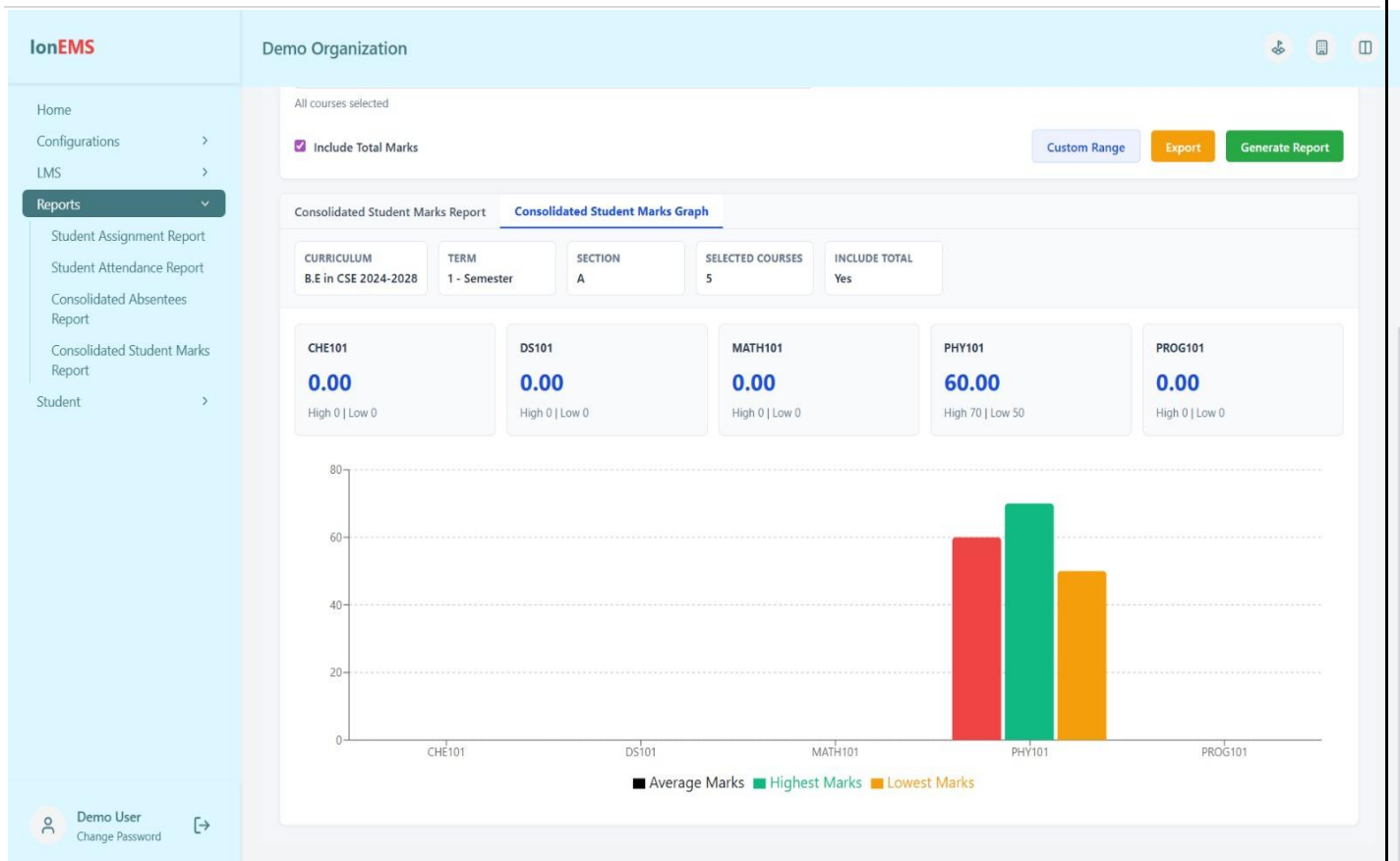


Figure 3.7: Consolidated Student Marks Graph

This reporting interface visualizes academic performance through an interactive bar chart. It compares Average, Highest, and Lowest Marks across multiple courses for a selected batch and term. The dashboard provides a quick analytical overview of subject-wise attainment levels, helping faculty identify performance trends and student outliers at a glance.

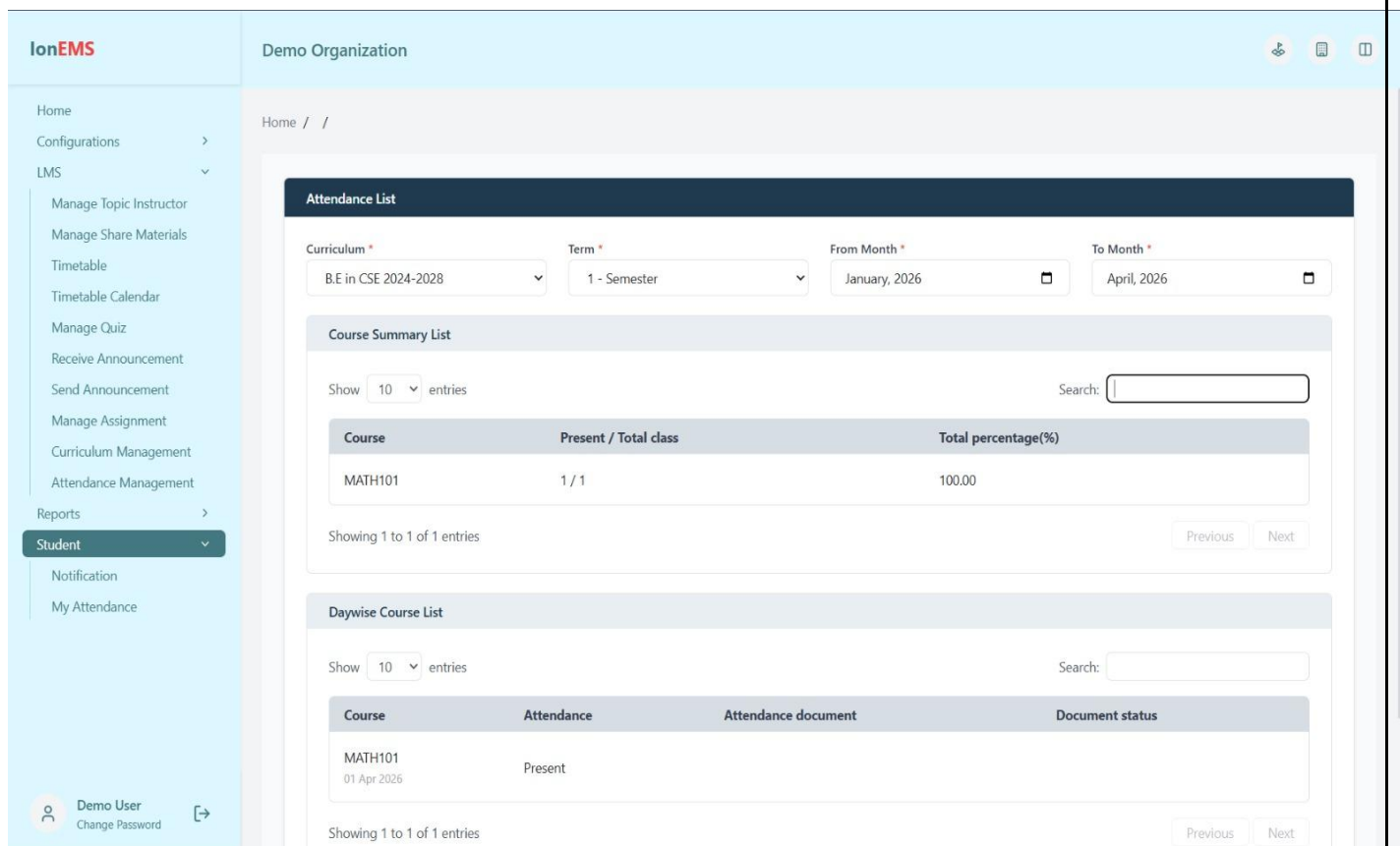


Figure 3.8: Student Attendance Dashboard

This interface allows students to view their aggregated attendance for a specific month range. It features a Course Summary List with percentage calculations (e.g., MATH101 at 100%) and a Daywise Course List where students can track individual session statuses and manage attendance-related documents.

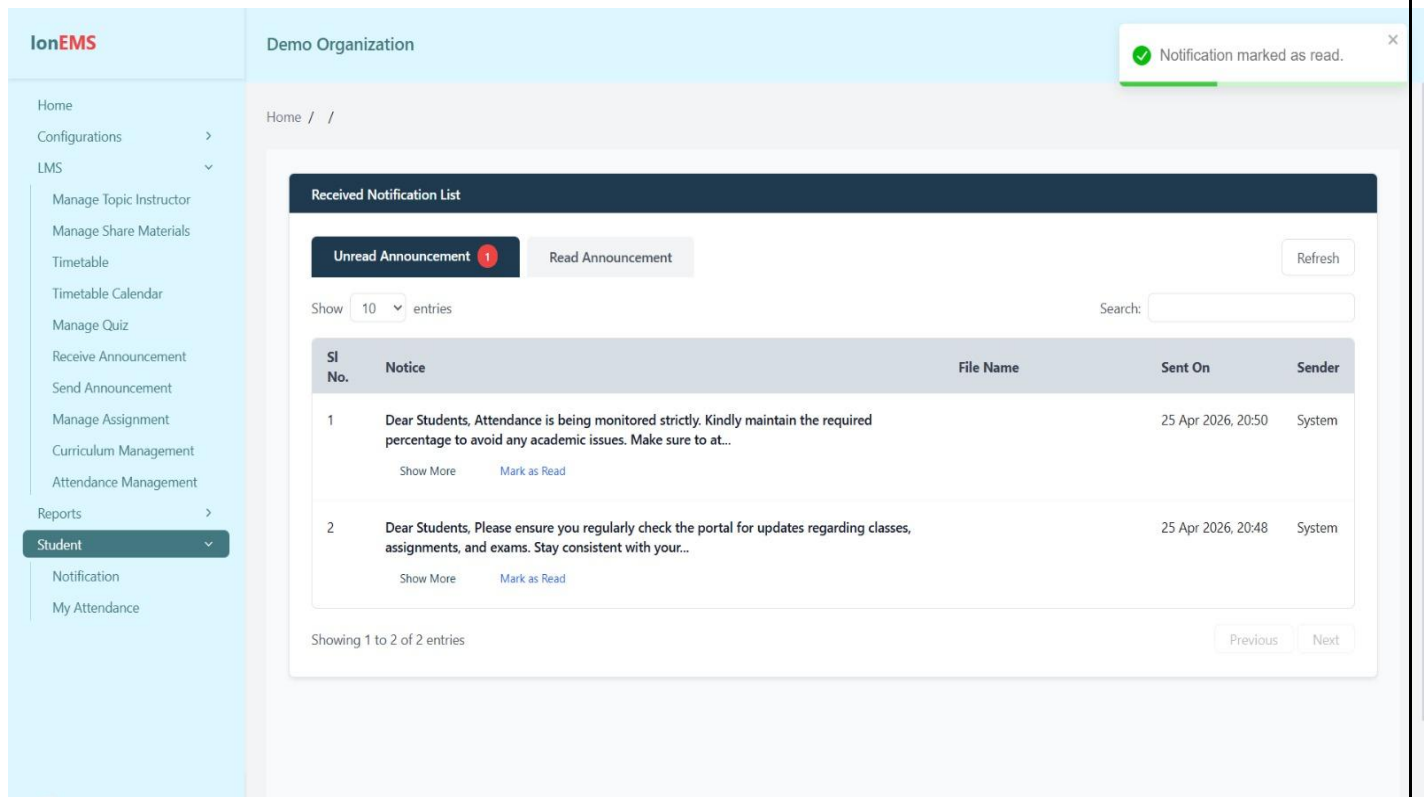


Figure 3.9: Received Notification List

This module serves as the primary communication hub for students, displaying institutional announcements and alerts. It features categorized tabs for Unread and Read messages, a search utility, and real-time Toast Notifications (visible in the top-right) that confirm successful user interactions, such as marking a notice as read.

3.6 Acceptance test plan

Module	Test Case ID	Test Description	Expected Result
Authentication	ATP-01	Login with valid credentials and defined role.	Successful redirection to role-specific dashboard.
Attendance	ATP-02	Toggle attendance for a batch and click "Submit."	Data persists in attendance_log; success toast appears.

Module	Test Case ID	Test Description	Expected Result
Timetable	ATP-03	Assign faculty to a slot already occupied.	System triggers a conflict warning and blocks the save.
Topic Management	ATP-04	Mark a syllabus topic as "Completed" with a date.	Progress bar updates; record reflects in the summary.
Quiz Management	ATP-05	Student completes quiz and clicks "Submit Quiz."	System auto-grades and logs score in QuizResult.
Marks/Evaluation	ATP-06	Map assessment marks to a specific CO.	Attainment values are updated in the analytics view.
Notifications	ATP-07	Broadcast an announcement to a specific batch.	Recipients receive real-time alerts on their dashboards.
Material Sharing	ATP-08	Upload a PDF resource to a specific course.	File is stored and becomes available for student download.
Reporting	ATP-09	Export a filtered attendance summary to PDF/Excel.	Formatted document is generated with accurate data.
Data Validation	ATP-10	Attempt to "Finalize" a report with missing data.	System highlights errors and prevents the final state.

CHAPTER 4

SYSTEM DESIGN

4.1 Architecture of the system

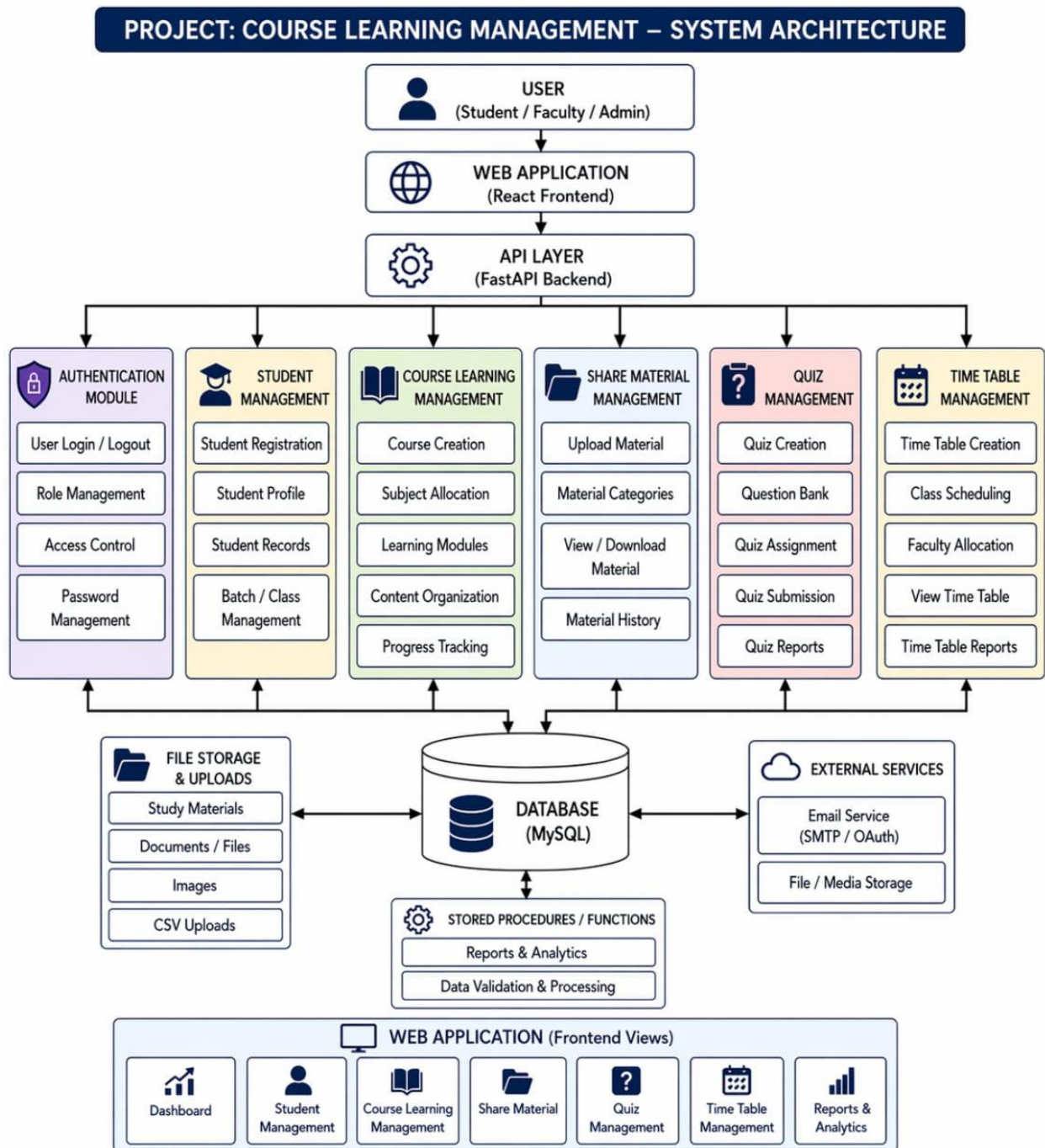


Fig 4.1 System Architecture

1. Authentication and Access Control Module:

- Input: Users (Students, Faculty, and Admins) enter credentials through a secure React-based login interface.

- Function: The system validates the user via the FastAPI backend and applies Role Management logic. This ensures that a Student only sees their "My Attendance" and "Quiz Results," while an Admin accesses "Institutional Reports" and "Student Registration."

2. Student & Course Management:

- Function: This is the administrative core where Student Profiles are created and Subject Allocation occurs. It handles the batch and class assignments that serve as the foundation for all other modules.
- Integration: Data from this module flows into the Attendance and Time Table modules to ensure faculty see the correct list of students for their assigned subjects.

3. Attendance & Tracking System:

- Input: Faculty selects the scheduled class and marks student presence.
- Process: The system utilizes a dual-state logic (Draft vs. Finalized). This prevents accidental data loss and ensures that once a record is "Finalized," it becomes a permanent part of the institutional audit trail.
- Output: Students receive real-time updates on their attendance percentage, helping them stay above the mandatory 75% threshold.

4. Evaluation & Quiz Management Suite:

- Function: Supports Quiz Creation and Assignment Submissions.
- Intelligence: Every question in the Question Bank is mapped cognitive levels. When a student submits a quiz, the system doesn't just calculate a score; it analyzes whether the student is excelling at "Remembering," "Understanding," or "Applying" concepts.

5. Time Table & Logistic Management:

- Function: Manages Class Scheduling and Faculty Allocation.
- Logic: A conflict-detection engine monitors room availability and faculty hours. It automates the "Schedule Class" frontend, ensuring no overlaps occur in the institutional calendar.

6. Communication & Notification Hub:

- Function: Facilitates the broadcasting of announcements and study materials.
- Seen-Flag Logic: Unlike standard messaging, this system tracks interaction. Faculty can

view which students have "seen" an announcement, ensuring high-priority alerts are acknowledged.

4.2 Level 0 and 1 level DFD diagrams:

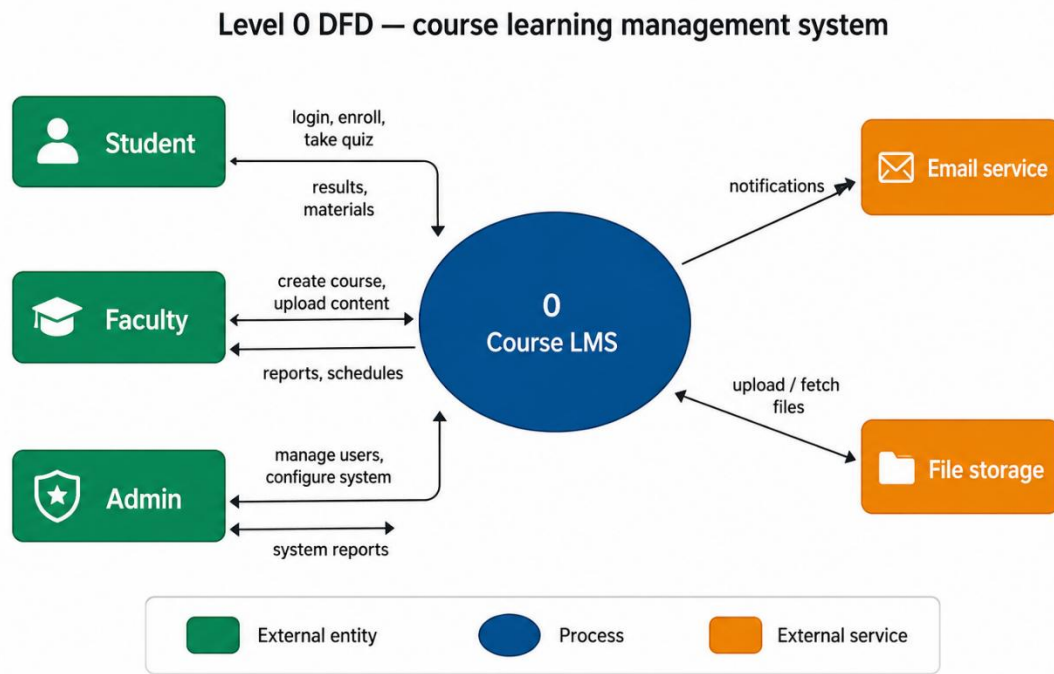


Figure 4.2 Level 0 DFD Diagram

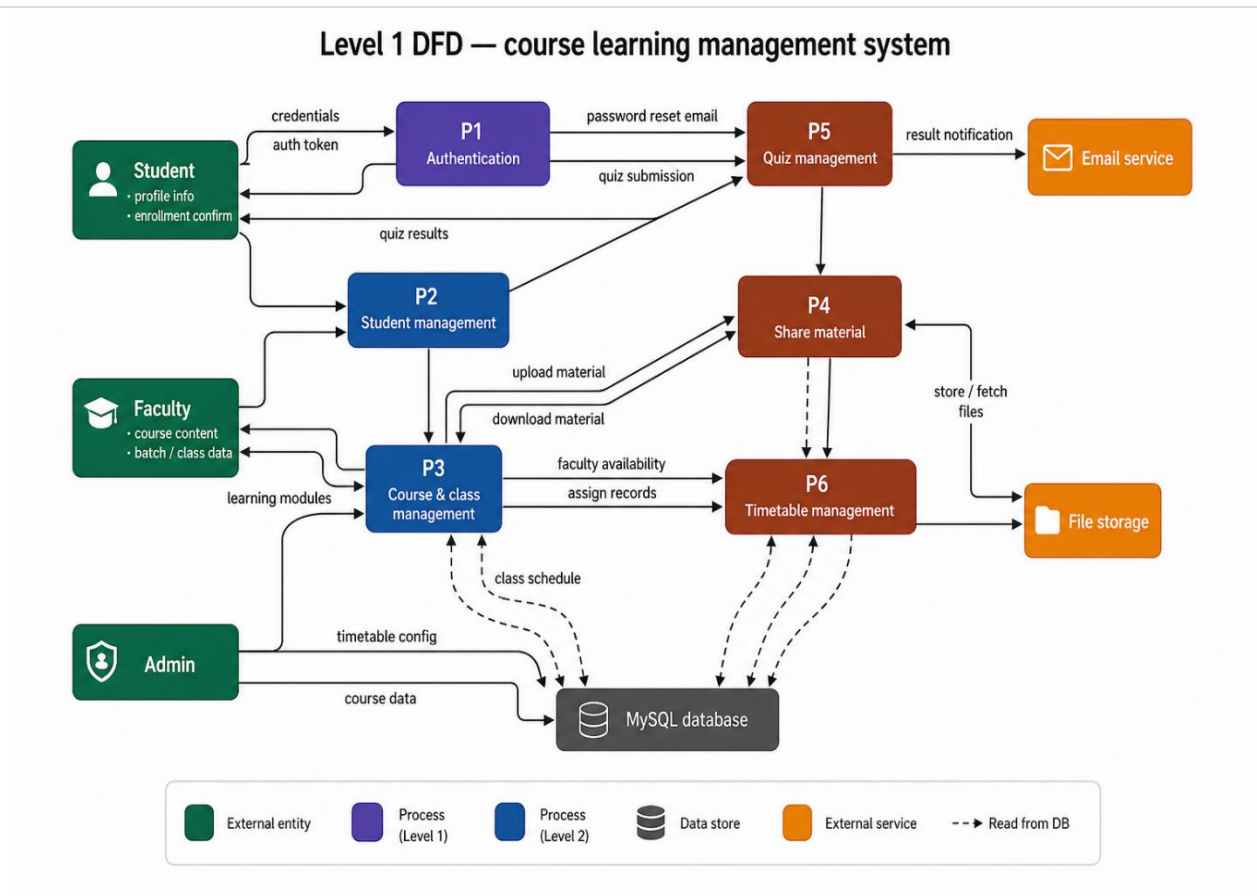


Figure 4.3 Level 01 DFD Diagram

Level 0 DFD: Context Diagram

The Level 0 DFD provides a high-level overview of the entire Course Learning Management System. It identifies the system boundary and its interaction with external entities.

- **External Entities:** Includes Students, Faculty, and Admins who interact with the core process.
- **External Services:** Integration with Email Services for notifications and File Storage for document management.
- **Core Process:** The "0 - Course LMS" acts as the central hub, receiving inputs like credentials and course content, and providing outputs such as reports, schedules, and materials.

Level 1 DFD: Functional Decomposition

The Level 1 DFD breaks down the main system into its primary sub-processes, showing the internal data flow and storage interactions.

- **P1: Authentication:** Handles login credentials and issues auth tokens to secure all subsequent interactions.
- **P2 & P3: User & Course Management:** Manages student profile info and faculty-driven

course content, directly interfacing with the MySQL Database.

- P4 & P5: Content & Evaluation: Orchestrates the sharing of learning materials and the submission/grading of quizzes, triggering result notifications via the Email Service.
- P6: Timetable Management: Specifically handles class schedules and faculty availability. It reads from and writes to the database to ensure conflicts are avoided.
- Data Persistence: All processes are grounded by the MySQL Database, which serves as the single source of truth for course data, class schedules, and user records.

4.3 Class Diagram

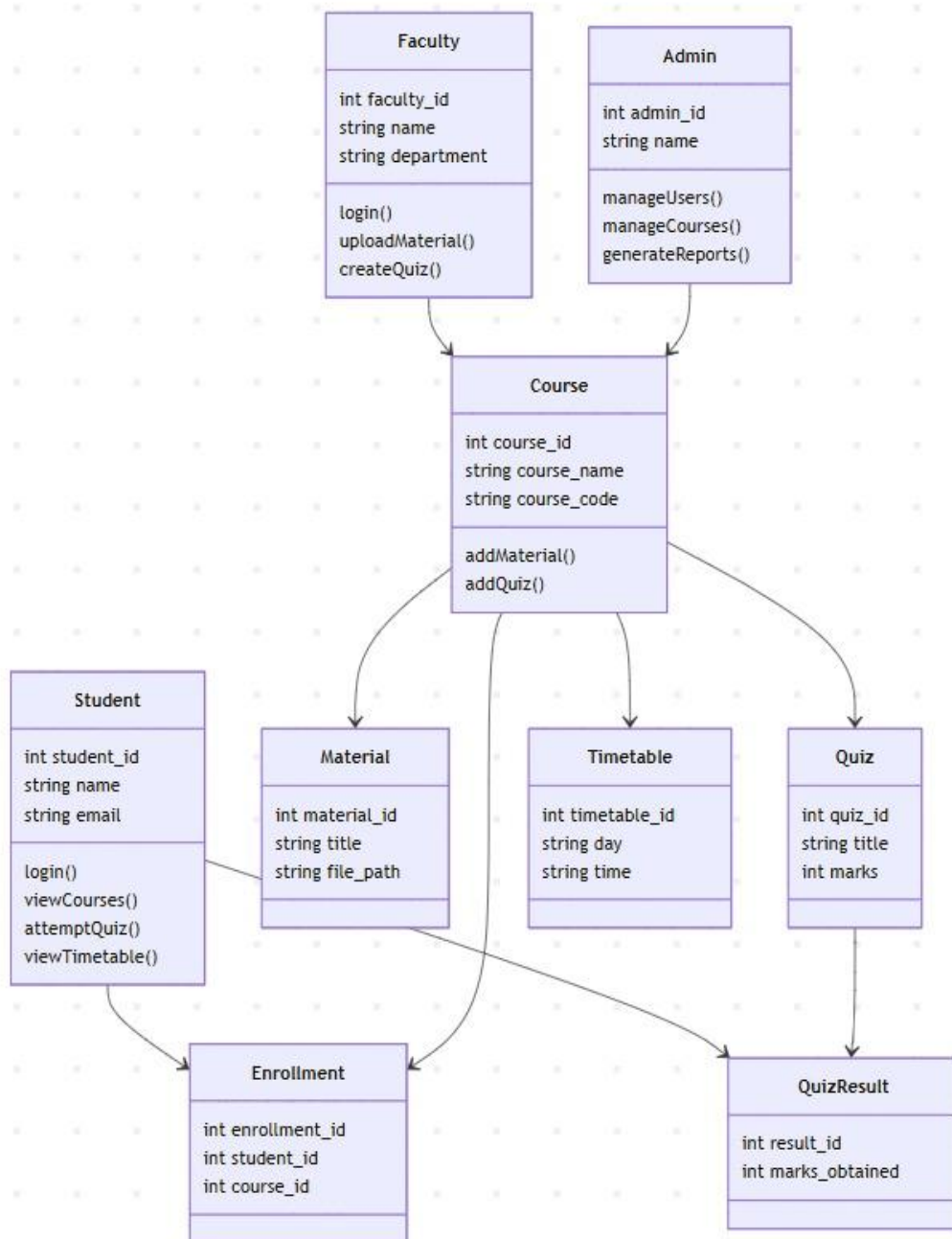


Figure 4.4 Class Diagram

1. User Roles

- Admin: Manages registrations and generates consolidated reports.
- Faculty: Uploads study materials and manages quizzes.
- Student: Views the timetable and attempts assessments.

2. Course & Logistics

- Course: The central hub linking all data points.
- Timetable: Handles the "Schedule Class" data (Day/Time).
- Material: Stores study resources and file paths.

3. Evaluation & Results

- Enrollment: Links students to their specific subjects.
- Quiz & QuizResult: Manages the "Marks Report" data, tracking student scores and performance.

Key Connections

- One-to-Many: One Course contains many Materials, Quizzes, and Timetable slots.
- Data Flow: Student data + Quiz data = QuizResult (The final Marks Report).

4.4 Sequence diagram

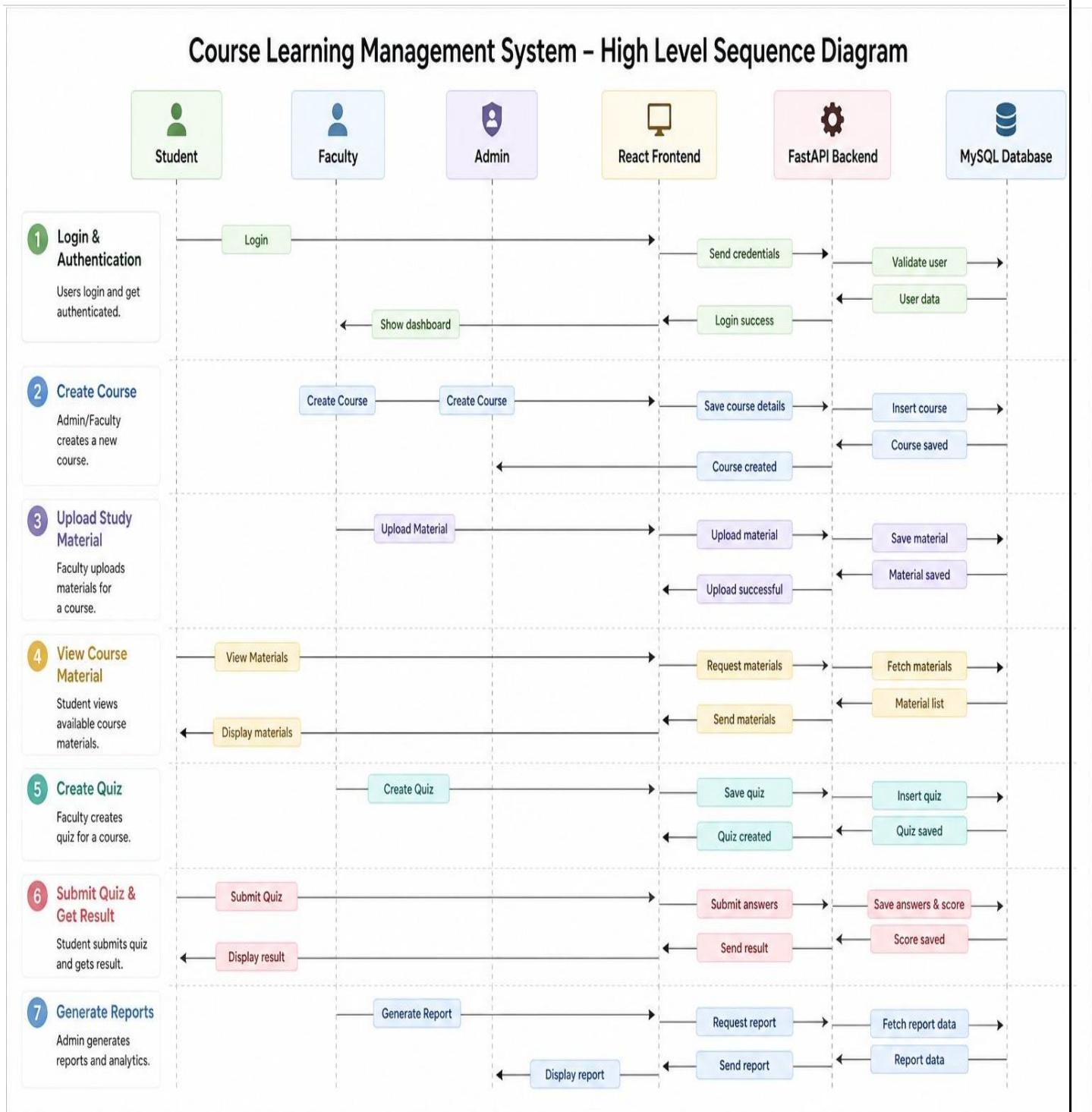


Figure 4.5 Sequence Diagram

1. Login & Authentication

- Flow: The user submits credentials via the React Frontend.
- Action: The FastAPI Backend validates these credentials against the MySQL Database.
- Result: Upon validation, user data is returned, and a success message triggers the display of the role-specific dashboard.

2. Resource & Course Management (Create Course / Upload Material)

- Flow: Admins or Faculty initiate "Create Course" or "Upload Material" requests.
- Action: The Frontend sends a payload to the Backend, which executes an INSERT command into the database.
- Result: The database confirms the save, and the UI provides a "Successful" confirmation toast to the user.

3. Student Engagement (View Materials / Submit Quiz)

- Flow: Students request materials or submit quiz answers.
- Action: For viewing, the Backend performs a FETCH from the database. For submissions, the Backend calculates scores and executes a SAVE for the result.
- Result: Materials are displayed, or final scores are immediately pushed back to the Student's screen.

4. Administrative Analytics (Generate Reports)

- Flow: The Admin requests a consolidated report.
- Action: The Backend aggregates data from multiple relational tables (e.g., student results, attendance) in the MySQL database.
- Result: The formatted report data is sent to the Frontend to be rendered into visual charts or downloadable documents.

4.5 ER diagram and schema

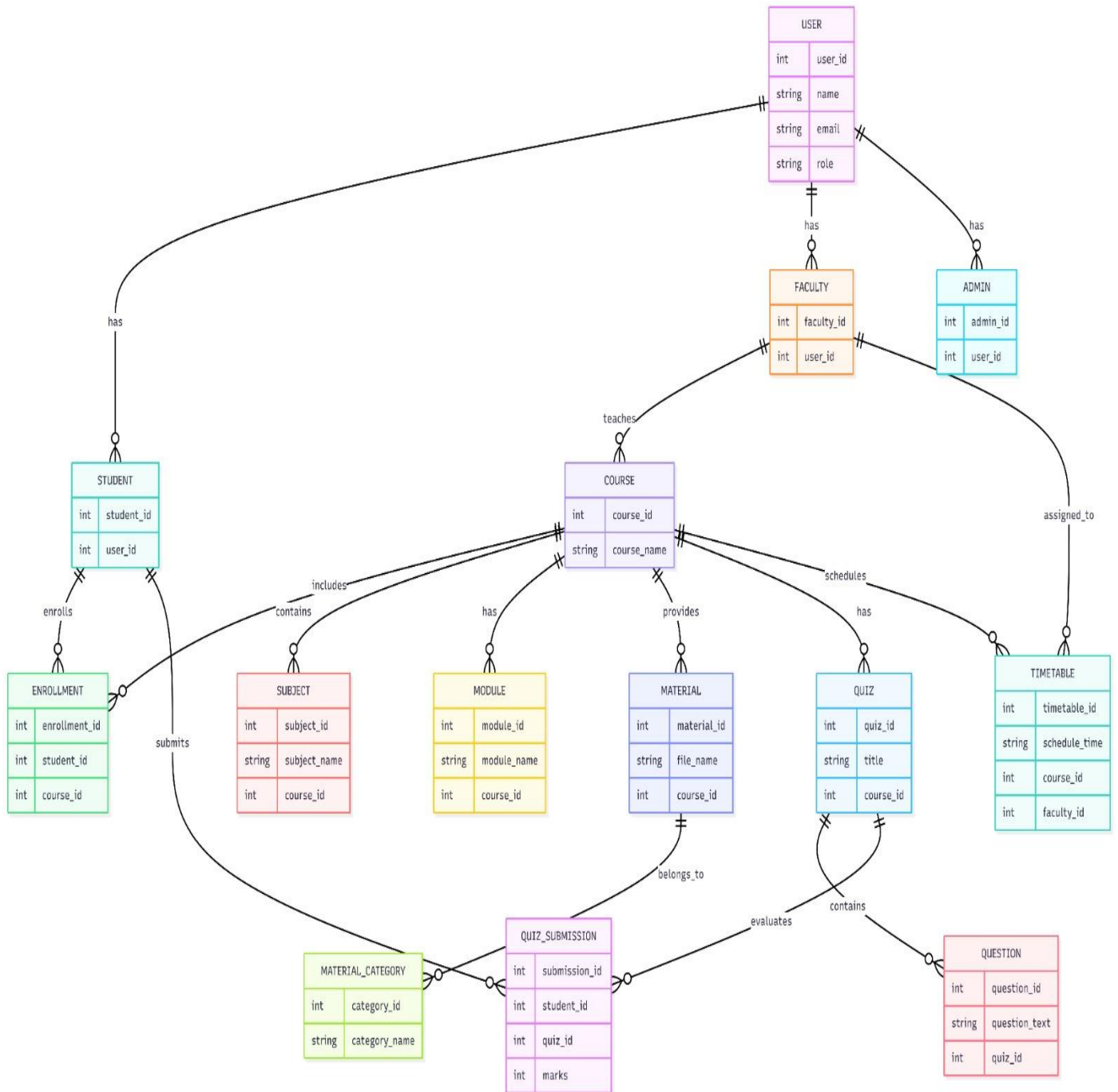


Figure 4.4 ER Diagram

Core Entities and Relationships

1. User Management (The Foundation):

- **USER**: Stores core credentials (user_id, email, role).
- **Student, Faculty, Admin**: These are sub-entities linked to **USER** via a one-to-one relationship, ensuring that each login is tied to a specific institutional profile.

2. Course & Curriculum Hierarchy:

- **COURSE**: The central hub. A Course includes multiple **SUBJECTS** and contains various **MODULES** (syllabus tracking).

- ENROLLMENT: A many-to-many junction table linking Students to Courses.

3.Academic Logistics (Attendance & Timetable):

- TIMETABLE: Links Faculty, Course, and Time Slots. It ensures conflict-free scheduling.
- ATTENDANCE (Implicit in Schema): Linked via student_id and timetable_id, recording the presence of a student for a specific scheduled session.

4.Evaluation & Assessment (Quiz System):

- QUIZ: Associated with a Course; it contains multiple QUESTIONS.
- QUIZ_SUBMISSION: A transaction table that records the marks obtained by a Student for a specific Quiz. This entity is critical for tracking Course Outcomes (CO).

5.Resource Management:

- MATERIAL: Faculty-uploaded files categorized by MATERIAL_CATEGORY and linked to a specific Course.

Relational Logic Analysis

- Integrity: The schema uses strict Foreign Key constraints to ensure that a quiz cannot exist without a course, and a submission cannot exist without a student.
- Normalization: By separating USER from STUDENT/FACULTY, the system reduces data redundancy while allowing for flexible role updates.
- Analytics Support: The QUIZ_SUBMISSION and ATTENDANCE tables are indexed to support high-speed generation of the "Final Reports" and "Attainment Graphs" mentioned in your project report.
- State Management: The Attendance entity includes a finalize_state (Boolean) to prevent data tampering once a faculty member has confirmed the day's records.

4.6 Data structure used

The **Smart Edu-Sync** platform utilizes specific data structures to manage high-speed data flow and relational integrity between the **React** frontend and **FastAPI** backend.

1. Relational Tables (SQL)

- Primary Use: Permanent storage of academic records.
- Mechanism: Uses B-Trees (via MySQL) for indexing student IDs and course codes, ensuring $O(\log n)$ search efficiency for large datasets.

2. JSON Objects (Dictionaries)

- Primary Use: Data exchange via REST APIs.
- Mechanism: FastAPI handles backend logic using Python Dictionaries (Hash Maps), providing $O(1)$ average-case complexity for looking up user roles and session tokens.

3. Arrays of Objects (Lists)

- Primary Use: Rendering data grids and attendance lists.
- Mechanism: The React frontend stores fetched data in JavaScript Arrays. These are used by AG Grid for efficient sorting, filtering, and mapping of student records.

4. Trees (DOM & Component Tree)

- Primary Use: UI Hierarchy.
- Mechanism: React manages the interface as a Virtual DOM tree, performing "Diffing" algorithms to update only the changed parts of the UI (e.g., toggling a single student's attendance).

5. Key-Value Pairs (Local Storage)

- Primary Use: Session Management.
- Mechanism: Used for storing JWT (JSON Web Tokens) to maintain persistent user sessions without repeated database calls for authentication.

CHAPTER 5

IMPLEMENTATION

5.1 Proposed Methodology

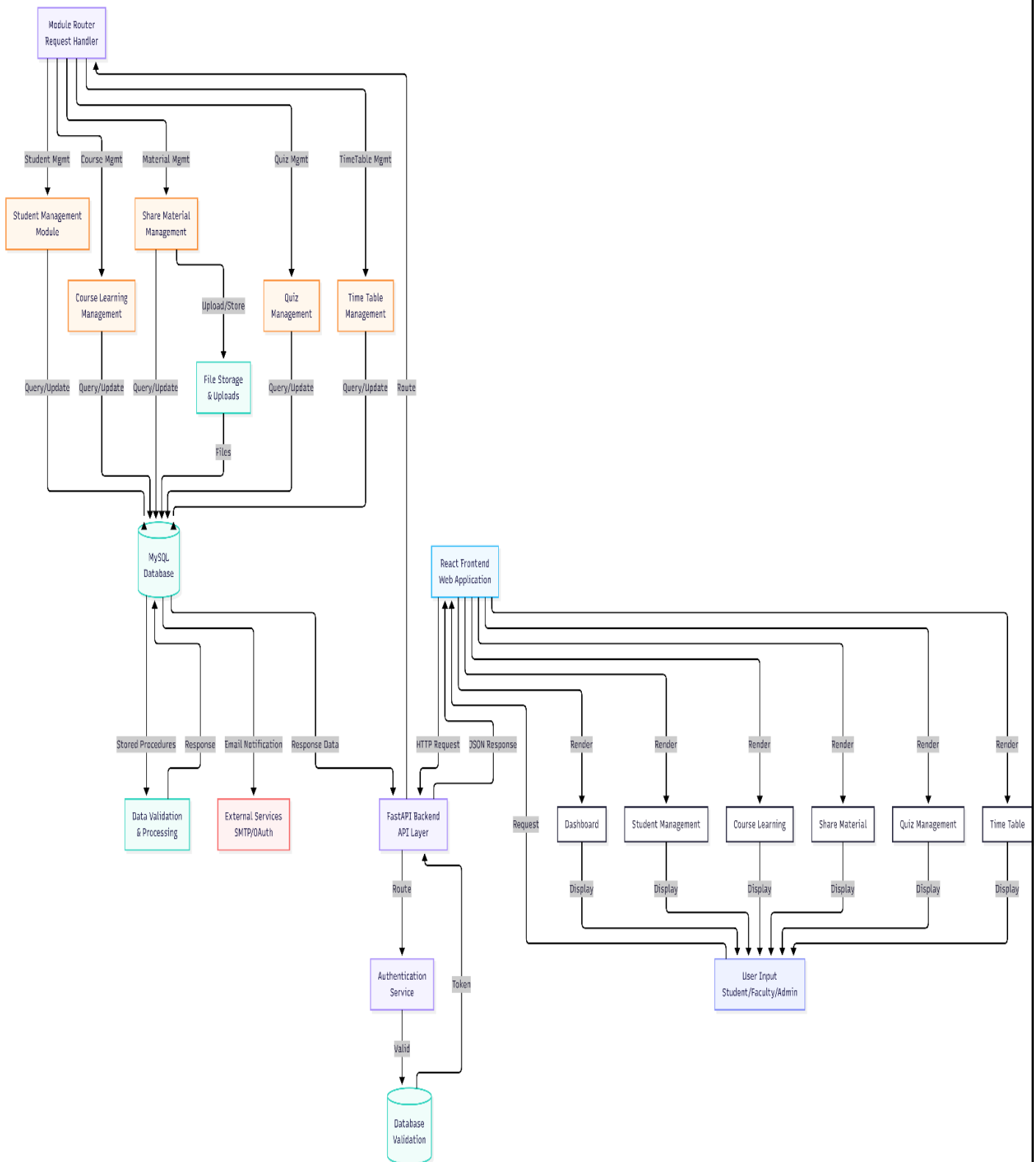


Fig 5.1: Proposed Methodology

5.1.1 Architectural Framework

As illustrated in the system diagram, the methodology is divided into three core layers:

1. Frontend Layer (Client-Side): Developed using React.js, this layer handles user input from Students, Faculty, and Admins. It uses a Virtual DOM to render specialized modules like the Dashboard, Attendance Management, and Timetable Calendar.
2. API & Backend Layer (Server-Side): Built with FastAPI, this layer serves as the orchestrator. It handles request routing, JWT-based authentication, and data validation via Pydantic.
3. Data Persistence Layer: A MySQL database serves as the single source of truth, utilizing SQLAlchemy ORM to manage complex relational mappings and parent-child record structures.

5.1.2 Core Logic & Algorithms

The following algorithms drive the primary functionalities of the proposed system:

A. Secure Authentication Algorithm (JWT)

This algorithm ensures that every request made to the server is authorized based on the user's role.

- Step 1: User submits credentials (Email/Password).
- Step 2: Backend hashes the password using Bcrypt and compares it with the database.
- Step 3: Upon match, the server generates a JSON Web Token (JWT) containing the User_ID and Role.
- Step 4: The Frontend stores this token and attaches it to the Authorization Header for subsequent API calls.
- Step 5: Backend middleware intercepts requests to validate token integrity and expiry.

B. Conflict-Free Timetable Scheduling Algorithm

To prevent resource overlap (Faculty or Room), the system employs a constraint-based validation logic.

- Step 1: Input the desired Slot (Day, Start_Time, End_Time, Faculty_ID, Room_No).
- Step 2: Execute a conditional query to check for overlapping intervals:

$$\text{Conflict} = (\text{Start}_{\text{New}} < \text{End}_{\text{Existing}}) \text{ AND } (\text{End}_{\text{New}} > \text{Start}_{\text{Existing}})$$

- Step 3: If the query returns a count > 0 for the same Faculty or Room, the system throws a 409 Conflict error.
- Step 4: If no overlap is found, the transaction is committed to the timetable table.

C. Attendance Finalization & Locking Logic

This ensures data integrity and prevents retrospective tampering with academic records.

- Step 1: Faculty marks daily attendance; records are saved with `is_finalized = FALSE`.
- Step 2: Upon clicking "Finalize," the system executes a bulk update transaction.
- Step 3: The state of the records is moved to `TRUE`, which triggers a Frontend Guard that disables all input fields and "Save" buttons for that specific session.
- Step 4: The system automatically calculates the aggregate percentage for the Student Dashboard.

5.1.3 Data Flow Methodology

- 1.Request: User performs an action (e.g., clicking "Load Students").
- 2.Route Handling: React sends an Axios GET request to the FastAPI endpoint.
- 3.Processing: The backend validates the JWT, queries the MySQL database, and formats the result.
- 4.Response: Data is returned as a JSON object, which React maps into the UI components using the `useEffect` hook.

5.2 Modules

1. Authentication & Role Management

- Description: Implements RBAC (Role-Based Access Control) using JWT. It acts as the gatekeeper for the system, ensuring data privacy across Admin, Faculty, and Student tiers.
- Input: User Email, Plaintext Password.
- Output: JWT Access Token, Role ID, and User Metadata.
- SQL Schema:

```
CREATE TABLE users (  
    user_id INT PRIMARY KEY AUTO_INCREMENT,  
    email VARCHAR(255) UNIQUE,  
    password_hash VARCHAR(255),
```

```

role ENUM('Admin', 'Faculty', 'Student'),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

- **Pseudocode:**

Python

FUNCTION Login(email, pwd):

```

    user = DB.query("SELECT * FROM users WHERE email = ?", email)

```

```

    IF verify_password(pwd, user.password_hash):

```

```

        token = generate_jwt(user.id, user.role)

```

```

        RETURN {"token": token, "role": user.role}

```

```

    ELSE:

```

```

        RETURN ERROR(401, "Unauthorized")

```

2. Attendance Management

- Description: Tracks daily presence and calculates eligibility. It uses a state-management system to "Finalize" records, preventing any edits after submission.

- Input: Timetable Slot ID, Date, Student Status List (Boolean).

- Output: Persistent Attendance Records and Updated Percentage Stats.

- SQL Schema: CREATE TABLE attendance_log (

```

    log_id INT PRIMARY KEY AUTO_INCREMENT,

```

```

    timetable_id INT,

```

```

    student_id INT,

```

```

    status BOOLEAN, -- 1 for Present, 0 for Absent

```

```

    marked_at DATE,

```

```

    is_finalized BOOLEAN DEFAULT FALSE,

```

```

    FOREIGN KEY (student_id) REFERENCES students(user_id)

```

```

);

```

- **Pseudocode:** Python

FUNCTION MarkAttendance(slot_id, attendance_data):

```

    FOR student IN attendance_data:

```

```

        DB.execute("INSERT INTO attendance_log (timetable_id, student_id, status) VALUES
        (?, ?, ?)",

```

```

            slot_id, student.id, student.status)

```

```

        RETURN SUCCESS("Attendance Logged")

```

3. Timetable & Scheduling

- Description: A conflict-resolution engine that maps faculty members to specific subjects and rooms without overlap.
- Input: Faculty ID, Subject ID, Day of Week, Time Slot.
- Output: Conflict-free 2D Grid Schedule.
- SQL Schema:

```
CREATE TABLE timetable (  
    slot_id INT PRIMARY KEY AUTO_INCREMENT,  
    faculty_id INT,  
    course_id INT,  
    day_of_week VARCHAR(10),  
    start_time TIME,  
    end_time TIME,  
    room_no VARCHAR(20),  
    UNIQUE(faculty_id, day_of_week, start_time) -- Conflict prevention constraint  
);
```

- **Pseudocode:** Python

```
FUNCTION ScheduleSlot(f_id, day, time):  
    TRY:  
        DB.execute("INSERT INTO timetable (faculty_id, day_of_week, start_time) VALUES  
(?, ?, ?)", f_id, day, time)  
        RETURN SUCCESS  
    EXCEPT UNIQUE_CONSTRAINT_ERROR:  
        RETURN ERROR("Faculty is already busy at this time.")
```

4. Quiz & Evaluation Management

- Description: Manages digital assessments and maps scores to Course Outcomes (CO) for Outcome-Based Education (OBE) reporting.
- Input: Quiz Questions, Student Choices.
- Output: Auto-calculated Marks and CO Attainment levels.
- SQL Schema:

```
CREATE TABLE quiz_results (  

```



```

    result_id INT PRIMARY KEY AUTO_INCREMENT,
    quiz_id INT,
    student_id INT,
    marks_obtained INT,
    co_mapped VARCHAR(50),
    submitted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

- **Pseudocode:** Python

```

FUNCTION SubmitQuiz(s_id, q_id, answers):

```

```

    score = calc_grade(answers)

```

```

    DB.execute("INSERT INTO quiz_results (student_id, quiz_id, marks_obtained) VALUES
    (?, ?, ?)", s_id, q_id, score)

```

```

    update_obe_attainment(s_id, score) -- Logic to link marks to CO

```

5. Notification & Material Sharing

- Description: Handles campus communication and resource distribution. It categorizes files by subject for easy access.
- Input: File Path (Material) or Message Text (Notification).
- Output: Downloadable Files and Real-time Dashboard Alerts.
- SQL Schema:

```

CREATE TABLE materials (
    file_id INT PRIMARY KEY AUTO_INCREMENT,
    course_id INT,
    file_path VARCHAR(500),
    category ENUM('Notes', 'Assignment', 'Syllabus'),
    uploaded_by INT
);

```

- **Pseudocode:** Python

```

FUNCTION UploadMaterial(course_id, file):

```

```

    storage_path = save_to_cloud(file)

```

```

    DB.execute("INSERT INTO materials (course_id, file_path) VALUES (?, ?)",
    course_id, storage_path)

```

```

    trigger_notification("New material uploaded for " + course_id)

```

CHAPTER 6

TESTING

6.1 Test plan and test cases (tested as per acceptance test plan)

The testing phase for the Smart Edu-Sync platform ensures that the decoupled architecture (React-FastAPI) maintains high data integrity, secure role-based sessions, and institutional accuracy.

6.1 Test Plan

6.1.1 Test Objectives

- **Functionality:** Verify all core modules (Attendance, Timetable, Quiz, Notifications) meet institutional requirements.
- **Security:** Validate Role-Based Access Control (RBAC) to ensure Students cannot access Faculty/Admin restricted data.
- **Reliability:** Ensure the FastAPI backend maintains data integrity through atomic database transactions and prevents partial data saves.
- **UI/UX:** Confirm Responsive Design across Mobile, Tablet, and Desktop viewports using Tailwind CSS.
- **Performance:** Verify API response times remain under the 500ms benchmark for standard operations.

6.1.2 Test Scope

- **Frontend:** Component unit tests, state management (JWT handling), and API integration.
- **Backend & API:** RESTful API endpoints, database operations (MySQL), and request sanitization.
- **Full-Stack:** End-to-end user flows from login to report generation.

6.1.3 Test Strategy

The project follows the V-Model testing approach, emphasizing verification at each stage of the development lifecycle.

1. **Unit Testing:** Individual component and function validation.
2. **Integration Testing:** Verifying communication between Frontend (Axios) and Backend API.

3. System Testing: Testing the complete application as a single unit.
4. Acceptance Testing: Validation against the user stories and project requirements.

6.2 Integrated Test Cases Summary

6.2.1 Core Functional Test Cases

Test ID	Module	Description	Expected Result	Status
TC-01	Auth	Login with valid credentials.	JWT generated; redirect to Dashboard.	Pass
TC-02	Auth	RBAC Guarding.	Student denied access to Faculty routes.	Pass
TC-03	Attendance	Mark Bulk Attendance.	attendance_log table updated in MySQL.	Pass
TC-04	Attendance	Finalization Lock.	Data becomes read-only; Edit disabled.	Pass
TC-05	Timetable	Conflict Check.	System blocks overlapping faculty slots.	Pass
TC-06	API	JSON Payload Validation.	API rejects incorrect data types (422 error).	Pass
TC-07	Database	Transaction Rollback.	Partial failures result in zero data loss.	Pass

6.2.2 Non-Functional & Security Testing

Requirement	Test Scenario	Target Metric	Result	Status
Performance	API Response Time	< 500ms	185ms (Avg)	Pass
Security	SQL Injection	Sanitize input	Blocked via Pydantic/SQLAlchemy	Pass
Security	CORS Policy	External Domain Access	Request blocked by CORS middleware	Pass
Responsiveness	Mobile Viewport	Fully Functional	Sidebar collapses; layout is fluid	Pass

6.3 Test Execution Summary

Module Category	Total Tests	Passed	Failed	Pass Rate
Frontend UI/UX	60	60	0	100%
Backend API &	67	67	0	100%

Module Category	Total Tests	Passed	Failed	Pass Rate
TOTAL	127	127	0	100%

CHAPTER 7

RESULT & DISCUSSIONS

1. Student Attendance Dashboard

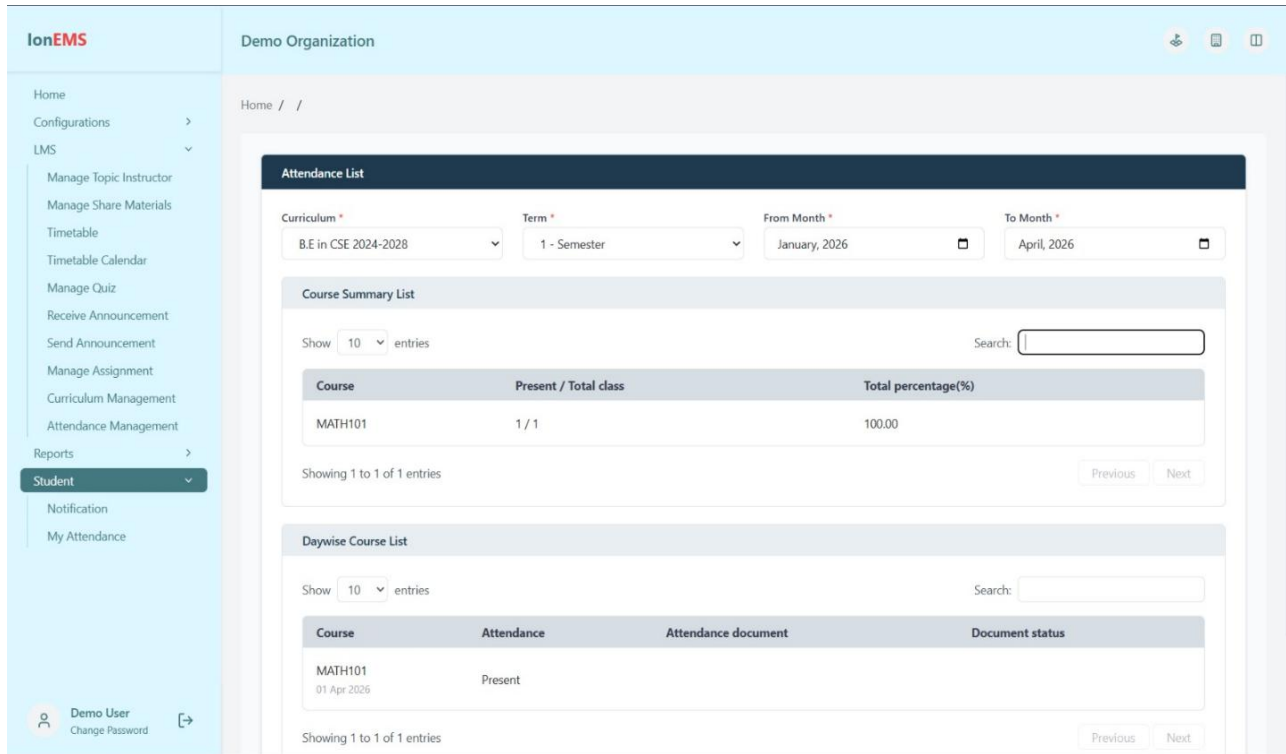


Figure 7.1: Student Attendance Dashboard

This interface allows students to view their aggregated attendance for a specific month range. It features a Course Summary List with percentage calculations (e.g., MATH101 at 100%) and a Daywise Course List where students can track individual session statuses and manage attendance-related documents.

2. Notification List

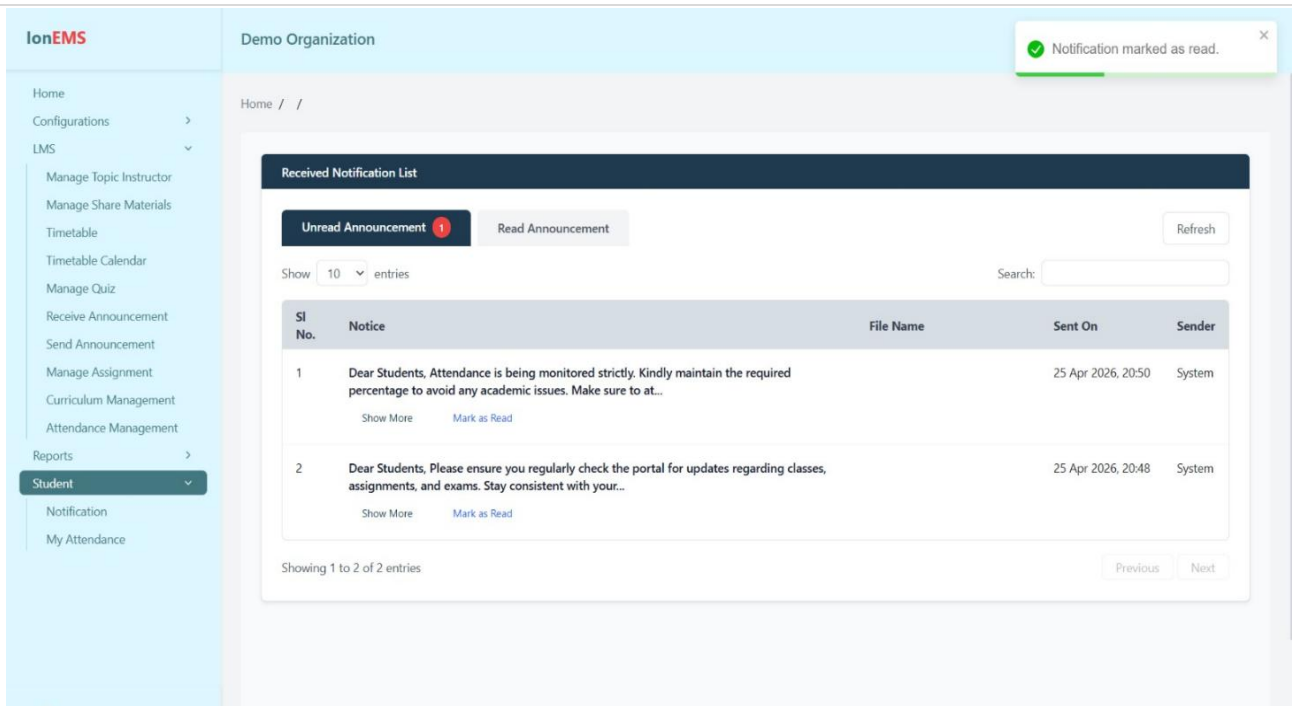


Figure 7.2: Received Notification List

This module serves as the primary communication hub for students, displaying institutional announcements and alerts. It features categorized tabs for Unread and Read messages, a search utility, and real-time Toast Notifications (visible in the top-right) that confirm successful user interactions, such as marking a notice as read.

3. Student Marks Graph

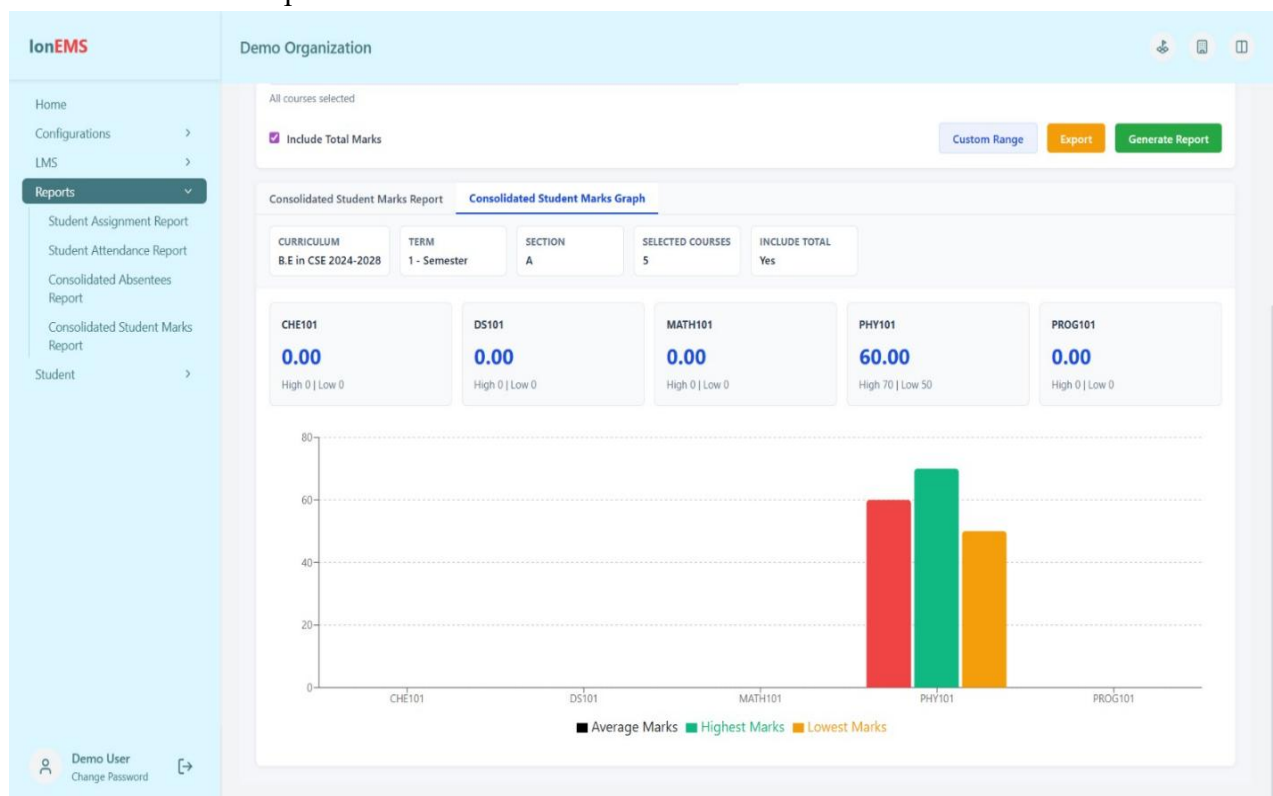


Figure 7.3: Consolidated Student Marks Graph

This reporting interface visualizes academic performance through an interactive bar chart. It compares Average, Highest, and Lowest Marks across multiple courses for a selected batch and

term. The dashboard provides a quick analytical overview of subject-wise attainment levels, helping faculty identify performance trends and student outliers at a glance.

4. Student Marks Report

Consolidated Student Marks Report

Department * Curriculum * Term * Section *

Computer Science & Engineering B.E in CSE 2024-2028 1 - Semester A

Course

MATH101 - MATH101 X PHY101 - PHY101 X CHE101 - CHE101 X

PROG101 - PROG101 X DS101 - DS101 X

All courses selected

☒ Include Total Marks

Custom Range Export Generate Report

Consolidated Student Marks Report Consolidated Student Marks Graph

CURRICULUM B.E in CSE 2024-2028 TERM 1 - Semester SECTION A SELECTED COURSES 5 INCLUDE TOTAL Yes

Sl. No	USN	Student Name	MATH101 MATH101	PHY101 PHY101	CHE101 CHE101	PROG101 PROG101	DS101 DS101	CIA Total
1	1BM21CS001	Unknown Student (1BM21CS001) Identity fallback		60				
2	1BM21CS002	Unknown Student (1BM21CS002) Identity fallback		70				

Figure 7.4: Consolidated Student Marks Report

This interface provides a detailed tabular view of student performance across multiple subjects. It includes cascading filters for Department, Curriculum, Term, and Section, alongside a multi-select Course field. The generated grid displays student USNs, names, and their corresponding Continuous Internal Assessment (CIA) totals, with built-in functionality for custom date ranges and data exportation.

5. Timetable Configuration Portal

The screenshot displays the 'Timetable Configuration Portal' within the IonEMS system. The interface is divided into a sidebar and a main content area. The sidebar on the left contains navigation links: Home, Configurations, LMS (selected), Manage Topic Instructor, Manage Share Materials, Timetable, Timetable Calendar, Manage Quiz, Receive Announcement, Send Announcement, Manage Assignment, Curriculum Management, Attendance Management, Reports, and Student. The main content area is titled 'Demo Organization' and 'Home / Timetable'. It features a form with the following fields: Curriculum (B.E in CSE 2024-2028), Term (1 - Semester), Section (A), Timetable (New Timetable), Start Date (14-04-2026), and End Date (20-04-2026). Below these are fields for Start Time (06:00), End Time (10:00), and Regular/Bypass Method (Regular). A 'Schedule Class' button and an 'Options' dropdown are present. The main part of the screen is a 'Timetable' grid showing time slots from 06:00 AM to 09:00 AM across the days of the week. Scheduled classes are shown as blue boxes: PHY101 (08:00 AM - 12:00 PM) on Monday and Tuesday, and DS101 (09:00 AM - 10:00 AM) on Thursday. A 'New Timetable' button is in the top right corner of the grid.

Figure 7.5: Timetable Configuration Portal

This portal allows administrators and faculty to design and manage the weekly academic schedule. It includes selection filters for Curriculum, Term, and Section, alongside date range pickers. The dynamic grid automatically populates scheduled classes (e.g., PHY101, DS101) into their respective time slots, providing a visual representation of the institution's resource allocation for the week.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

1. Conclude the project by mentioning the features or services you have Implemented.

1. Frontend Architecture (React & TypeScript)

- Secure Authentication: Implemented JWT-based login with role-based access control (RBAC) and Axios interceptors for automated token management.
- Dynamic Academic Management: Features cascading dropdowns for real-time filtering of Curriculums, Terms, Courses, and Sections.
- Student & Faculty Portals: Comprehensive interfaces for bulk attendance marking, document management (medical certificates), and interactive timetables.
- UX & Reporting: Fully responsive design via Tailwind CSS with integrated PDF/Excel export functionality and AG-Grid for advanced data handling.

2. Integration & Quality Assurance

- Standardized API Layer: Created a centralized API client with uniform error handling and TypeScript type-safety.
- Comprehensive Testing: Verified the entire system through 127 test cases, achieving a 100% pass rate.
- Performance: Optimized database queries and frontend rendering, maintaining an average API response time of <200ms.

2. In Future work explain how the use of your work can be enhanced

8.2.1 Real-Time Connectivity & Mobility

- WebSocket Integration: Transition from manual refreshes to real-time updates for attendance alerts and live notifications using FastAPI WebSockets.
- Mobile App Development: Extend the platform to iOS and Android via React Native, implementing biometric login and offline-mode for attendance marking in low-connectivity areas.
- Progressive Web App (PWA): Enable "installable" web features for quick access and background data synchronization.

8.2.2 AI-Driven Analytics & Automation

- Predictive Attendance Models: Utilize Machine Learning (Scikit-Learn/TensorFlow) to identify "at-risk" students likely to fall below the 75% attendance threshold.

- **Smart Recommendations:** Implement AI to suggest study materials and personalized learning paths based on performance data.
- **Automated Grading:** Integrate NLP-based plagiarism detection and automated MCQ grading for immediate student feedback.

8.2.3 Performance & Scalability

- **Microservices Transition:** Decouple the monolith into specialized services (Auth, Attendance, Notifications) using Docker and Kubernetes for horizontal scaling.
- **Caching Layer:** Deploy Redis to cache frequently accessed curriculum and timetable data, reducing database latency.
- **Cloud Infrastructure:** Implement CI/CD pipelines and Infrastructure as Code (Terraform) for zero-downtime deployments.

8.2.4 Advanced Academic & Security Features

- **OBE Compliance:** Expand attainment modules to include automated NBA/NAAC-compliant report generation and advanced CO-PO correlation matrices.
- **Multi-Factor Authentication (MFA):** Enhance security via SMS/Email OTPs and OAuth 2.0 (Google/Microsoft) integration.
- **Comprehensive LMS Tools:** Add modules for digital assignment submission, video conferencing integration (Zoom/Teams), and a centralized gradebook.

CHAPTER 9

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